

OISD STANDARD - 137

**FOR RESTRICTED
CIRCULATION ONLY**

**STANDARD
FOR
INSPECTION OF ELECTRICAL EQUIPMENT**

OISD STANDARD - 137

First Edition, January 1990
Reaffirmed, August, 1999
Amended Edition, October, 2010
Revised Edition, April, 2016



Oil Industry Safety Directorate

Government of India
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Prepared by

**FUNCTIONAL COMMITTEE ON
INSPECTION OF ELECTRICAL EQUIPMENT**

OIL INDUSTRY SAFETY DIRECTORATE

Government of India

Ministry of Petroleum & Natural Gas

8th Floor, OIDB Bhavan, Plot No. 2, Sector – 73, Noida – 201301 (U.P.)

Preamble

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for sovereignty and prosperity of the country. As the industry essentially deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as only safe performance at all times can ensure optimum ROI of these national assets and resources including sustainability.

While statutory organizations were in place all along to oversee safety aspects of Indian petroleum industry, Oil Industry Safety Directorate (OISD) was set up in 1986 by Ministry of Petroleum and Natural Gas, Government of India as a knowledge centre for formulation of constantly updated world-scale standards for design, layout and operation of various equipment, facility and activities involved in this industry. Moreover, OISD was also given responsibility of monitoring implementation status of these standards through safety audits.

In more than 25 years of its existence, OISD has developed a rigorous, multi-layer, iterative and participative process of development of standards – starting with research by in-house experts and iterating through seeking & validating inputs from all stake-holders – operators, designers, national level knowledge authorities and public at large – with a feedback loop of constant updation based on ground level experience obtained through audits, incident analysis and environment scanning.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry culminating in a safer environment in the industry. OISD – except in the Upstream Petroleum Sector – is still a regulatory (and not a statutory) body but that has not affected implementation of the OISD standards. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption in various statutory rules of the land.

Petroleum industry in India is significantly globalized at present in terms of technology content requiring its operation to keep pace with the relevant world scale standards & practices. This matches the OISD philosophy of continuous improvement keeping pace with the global developments in its target environment. To this end, OISD keeps track of changes through participation as member in large number of International and national level Knowledge Organizations – both in the field of standard development and implementation & monitoring in addition to updation of internal knowledge base through continuous research and application surveillance, thereby ensuring that this OISD Standard, along with all other extant ones, remains relevant, updated and effective on a real time basis in the applicable areas.

Together we strive to achieve NIL incidents in the entire Hydrocarbon Value Chain. This, besides other issues, calls for total engagement from all levels of the stake holder organizations, which we, at OISD, fervently look forward to.

Jai Hind!!!

Executive Director

Oil Industry Safety Directorate

FOREWORD

The Oil Industry in India is 100 years old. As such variety of practices has been in vogue because of collaboration / association with different foreign companies and governments. Standardisation in design philosophies and operating and maintenance practices at a national level was hardly in existence. This coupled with feedback from some serious accidents that occurred in the recent past in India and abroad, emphasized the need for the industry to review the existing state of art in designing, operating and maintaining oil and gas installations.

With this in view, the Ministry of Petroleum & Natural Gas, in 1986, constituted a Safety Council assisted by Oil Industry Safety Directorate (OISD), staffed from within the industry, in formulating and implementing a series of self-regulatory measures aimed at removing obsolescence, standardising and upgrading the existing standards to ensure safe operations. Accordingly, OISD constituted a number of Functional Committees of experts nominated from the industry to draw up standards and guidelines on various subjects.

The present document on “Inspection of Electrical Equipment” was prepared by the Functional Committee on “Inspection of Electrical Equipment. This document is based on the accumulated knowledge and experience of Industry members and the various national and international codes and practices. It is hoped that provisions of this standard if implemented objectively, may go a long way to improve the safety and reduce accidents in Oil and Gas Industry. Users of this document are cautioned that no standard can be a substitute for the judgement of responsible and experienced engineer.

Suggestions for amendments, if any, to this standard should be addressed to:

The Co-ordinator,
Committee on
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These documents are intended only to supplement and not replace the prevailing statutory requirements.

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INSPECTION OF ELECTRICAL EQUIPMENT

2016

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INSPECTION OF ELECTRICAL EQUIPMENT

CONTENTS

SECTION	DESCRIPTION
1.0	Introduction
2.0	Scope
3.0	Definitions
4.0	Permit-to-Work (Electrical)
5.0	Basis and Types of Inspection
6.0	Frequency of Inspection
6.1	Rotating Machinery
6.1.1	Generator
6.1.2	Motor
6.2	Stationary Equipment
6.2.1	Switchboard & Panel
6.2.2	Transformer, Reactor & Bus duct
6.2.3	Battery
6.2.4	Battery Charger & D.C. board
6.2.5	VFD/ UPS/ Inverter
6.2.6	Capacitor Bank
6.2.7	Switchyard Equipment
6.2.8	Lighting Fixture & Junction Box
6.2.9	Test Instruments for Inspection
6.3	Protective Systems
6.3.1	Relay & Meter
6.3.2	Earthing & Bonding
6.3.3	Lightning Protection
6.3.4	Flameproof/ Weather protection
6.3.5	Fire Alarm System
6.3.6	Cathodic Protection

6.3.7	Electric Heat Tracing
6.3.8	Protective & Safety appliances
6.4	Distribution Network
6.4.1	Cable and Conduit wiring
6.4.2	Cable Duct/ Trench/ Tray support
6.4.3	Overhead Line (Uninsulated)
6.5	Outdoor Switchyard
6.6	Switchgear Room/ Substation Building
6.7	Safe Procedures & Systems
7.0	Procedure for electrical tests & recommended values
8.0	Documentation
9.0	Precommissioning checks
10.0	References

ANNEXURES

- I. History Cards
- II. Failure Analysis Report
- III. Details of Designated Personnel
- IV. Cable Fault Detection
- V. Typical Diagrams of Resistances to general mass of earth

	OISD – STD – 137	Page No. 1
	INSPECTION OF ELECTRICAL EQUIPMENT	

INSPECTION OF ELECTRICAL EQUIPMENT

1.0 INTRODUCTION

Electricity constitutes one of the major sources of ignition for fire accidents and explosions in petroleum installations where hazardous atmosphere is normally encountered. In so far fire losses due to electricity are concerned, it is estimated to account for approximately 30% of all fire losses. Besides equipment damage and property loss, electrical hazards also results in injuries and fatalities to personnel due to electric shock.

Timely inspection, condition monitoring and preventive maintenance of electrical equipment and its connected systems will go a long way in ensuring safer operations, for a longer period of time, of the electrical installations in the Oil Industry.

2.0 SCOPE

This standard specifies the minimum inspection requirements for the installed electrical systems in the Oil Industry, during their operation and maintenance with special reference to the safety aspects. Also included are:

- i) Permit-to-work (Electrical) or Electrical Line Clearances covering special safety requirements for jobs requiring Electrical Isolation/ Lockout.
- ii) Pre-commissioning check items for new installations as well as for repaired equipment.

This standard in no way supersedes the statutory regulations of the Chief Electrical Inspectorate, Factory Inspectorate, Petroleum and Explosives Safety Organisation, Directorate General of Mines Safety, or other government bodies, which shall be followed, as applicable.

3.0 DEFINITIONS

i) DESIGNATED ELECTRICAL PERSON

Designated Electrical Person (s) is / are the responsible person (s) of the Electrical Section in the owner organisation, designated by the organisation to carry out Electrical Isolation / Energisation. He is also designated under the provisions of CEA Regulation 19(3), 37 (i), 44 (1)(i), 44(2)(vii), 44(2)(viii x, xi), 94(2), 105 (1)(2), 105(5), 108 and 109(8).

Note:

1. Owner organisation shall maintain a list of such Designated Electrical person as per CEA regulations clause 3(2)
2. Suggestive format for details of designated personnel available in Annexure-III.

ii) ENGINEER-IN-CHARGE (EIC)

EIC is competent person of the owner organisation, in charge of the work, to whom work permit shall be issued by Unit-in-charge. In case the work involves electrical jobs, EIC shall be an electrical engineer.

EIC shall be responsible for continuous supervision of the work and for arranging safe Isolation/Energisation of the electrical equipment through designated Person as detailed in Section 4.0 to ensure safety while working on the equipment and associated systems. He

	OISD – STD – 137	Page No. 2
	INSPECTION OF ELECTRICAL EQUIPMENT	

shall also ensure the implementation of effective safety measures by personnel/ agencies working under his supervision, throughout the duration of work.

iii) UNIT-IN-CHARGE (UIC)

UIC is Owner-in-charge as defined in OISD-STD-105 (Work Permit System) authorised to issue Work Permit to work in the respective Unit under his jurisdiction (e.g. Chief Chemist shall be UIC for Laboratory).

iv) ELECTRICAL-IN-CHARGE (ELIC)

Electrical-in-charge is a competent person in charge of the Electrical Department/Section of the owner organisation responsible for safe operation and maintenance of all electrical systems. The system of issuing Electrical Line Clearances as detailed in Section 4.0 and conformity of installations with statutory requirements from time to time, shall also be ensured by him.

v) ELECTRICAL SAFETY OFFICER

Electrical Safety Officer is a competent person in the Electrical discipline of the owner organisation who shall ensure observance of safety measures specified under CEA Safety Regulation 2010 in their organisation for construction, operation and maintenance of power stations, sub-stations, transmission and distribution lines. He shall possess the competence required for Electrical Safety Officer as per CEA Regulation 5.

vi) PERMIT-ISSUING AUTHORITY (ELECTRICAL)

Permit-Issuing Authority (Electrical) is a competent person in the Electrical Department/Section of the Owner organisation, designated by Electrical in-charge to issue Permits (Electrical Line Clearances) and related safety instructions. He shall issue Permit only after ensuring that required competent electrical staff, depending upon the specific nature of work, are actually engaged in the work.

vii) ELECTRICAL LINE CLEARANCES:

These Clearances are electrical safety requirements related to jobs covered under Permit-to work(Electrical) as described in Section 4.0

Both the terms “Permit-to work (Electrical)” and “Electrical Line Clearance” (briefly “Permit” or “Clearance”) referred in this Standard convey the same meaning and are interchangeable.

viii) LOCKOUT/TAGOUT (LOTO):

The placement of a lockout and tagout devices on an energy isolating device, in accordance with an established procedure, to ensure and indicate that the energy isolating device and the equipment being controlled cannot be operated until the lockout and tagout devices are removed.

ix) ENERGISED

Energised or ‘live’ means electrically charged.

x) ISOLATED

Isolated or ‘dead’ means at or about earth potential and disconnected from any live system. It is used only with reference to current carrying parts when these parts are not live.

	OISD – STD – 137	Page No. 3
	INSPECTION OF ELECTRICAL EQUIPMENT	

4.0 PERMIT-TO-WORK (ELECTRICAL) OR ELECTRICAL LINE CLEARANCES

4.1 SCOPE

This section shall be read in conjunction with the OISD STD-105 (Work Permit System). The scope in this section covers the special requirements for any job on electrical system where Electrical Isolation-Lockout/Energisation or Earth Excavation Clearances are also needed.

For definitions of terms and responsibilities of various categories of staff involved in Permit System, Section 3.0 shall be referred.

4.2 APPLICATION

For any construction, repair, testing, maintenance or inspection jobs on an electrical system or for any work in close proximity to energised electrical circuits, or for that matter any job in general where an Electrical Isolation is needed, the requirements detailed in this section shall apply.

Compliance to these requirements is necessary to prevent:

- i) The equipment/line becoming accidentally or inadvertently electrically charged when persons are working on them (as cautioned in CEA Regulation 19).
- ii) Induction effect from the nearby energised electrical systems.
- iii) Accidental contact by the personnel with any live part of electrical circuit.

The requirements detailed in Section 4.6 shall apply for obtaining clearances during earth excavation.

4.3 TYPES OF CLEARANCES

Following Electrical Line Clearances, through a written Permit-to-work system, shall be obtained by EIC from Permit-Issuing Authority (Electrical), for any work covered under Permit system as detailed in this Section, in addition to Work Permit from UIC.

- i) Electrical Isolation-Lockout Line Clearance (before commencement of work) and Electrical Energisation Line Clearance before energisation of isolated equipment and circuit (after completion of work).
- ii) Earth Excavation Clearance.
- iii) In special cases, when Checking/ Testing/ maintenance job (like control circuit checking, measurement, interlock checking etc.) are to be carried out on equipment/ circuit which are energized (live), specific safety related work practices, in consistent with the nature & extent of the associated electrical hazards, shall be employed :
 - 1) All Work Permits (except Work Permit issued to EIC for job on energized equipment/ circuit), pertaining to that particular equipment and associated facilities, shall be withdrawn by UIC during such checking.
 - 2) Safe working procedure, suitable for the conditions under which work is to be performed, shall be determined.
 - 3) Time period for the job to be clearly specified.
 - 4) Job list / checklist for the jobs to be kept ready.
 - 5) Job Safety analysis for the job to be carried out by UIC and EIC.
 - 6) Rough sketch, if necessary, should be attached depending upon hazard.

	OISD – STD – 137	Page No. 4
	INSPECTION OF ELECTRICAL EQUIPMENT	

4.4 PROCEDURE FOR ELECTRICAL ISOLATION - LOCKOUT / ENERGISATION LINE CLEARANCES

Whenever the jobs covered under Permit system fall into any of the following four categories, the procedure detailed below (applicable to the respective category) and Special Instructions as per section 4.5 shall be followed. Electrical Isolation and Energisation shall be carried out only by Designated Person.

4.4.1 UIC requiring an electrical system to get checked through Electrical-in-charge (e.g. An electric motor trips repeatedly interrupting plant operation).

- i) UIC issues Job Request along with Work Permit to Electrical in charge to carry out the work
- ii) Designated person after receiving Work Permit does the positive Electrical Isolation (If required), completes all the work, makes the circuits ready for energisation, reports to Electrical-in-charge, and returns Work Permit.
- iii) Electrical-in-charge verifies the completion of all work, and returns Work Permit to UIC, after ensuring that the circuits are kept ready for safe re-energisation.
- iv) UIC decides when to start the equipment for operation.

4.4.2 Electrical-in-charge requiring to complete departmental job of checks/repair/test on an electrical system.

- i) EIC raises job request specifying the job requirement to UIC
- ii) UIC issues Work Permit duly authorised along with Electrical Isolation Line clearance to EIC for carrying out the specific work, after getting clearance from production-in-charge.
- iii) EIC on completion of the work returns Work Permit and Clearance to UIC, after ensuring that the circuits are kept ready for safe re-energisation
- iv) UIC decides when to start the equipment for operation and arranges for safe charging of the equipment, after getting clearance from production-in-charge.

4.4.3 Another Department requiring execution of a job on an electrical system (e.g. Revamping work or a design improvement etc.)

- i) Job Request by concerned Department Engineer in charge (who proposes to carry out the specific work) is raised to UIC with details of work. Rough sketch, if necessary, should be included depending upon hazard.
- ii) UIC requests Electrical-in-charge in writing for positive Electrical Isolation.
- iii) Designated person does electrical isolation, and submit the electrical isolation line clearance to UIC.
- iv) UIC issues Work Permit duly authorised to EIC for carrying out the specific work.
- v) EIC on completion of the work returns Work Permit to UIC along with the document showing the modification being carried out, after ensuring that the circuits are kept ready for safe re-energisation.

	OISD – STD – 137	Page No. 5
	INSPECTION OF ELECTRICAL EQUIPMENT	

- vi) UIC decides when to start the equipment for operation and arranges for safe charging of the equipment.

4.4.4 Another Department requiring execution of a job on a non-electrical equipment (e.g. Pump, compressor, fan, etc.) having electric drive.

Procedure identical to section 4.4.3 shall be followed.

4.5 SPECIAL INSTRUCTIONS

- i) The system of issuing Work Request, Electrical Line clearances (for Isolation-Lockout/Energisation) and Work Permit, shall be implemented through a written Permit-work, system in the prescribed format which shall be developed by individual organisation depending on the type of installation and administrative set up, taking care that the provisions of CEA Regulations 3, 5, 12, 19; IS:5216 and the Owner-in-charge concept, as defined in OISD-STD-105 (Work Permit System) are incorporated in this format.
- ii) Authorised Person while carrying out the work covered under Permit system shall physically possess Permit duly authorised by Permit-Issuing Authority (Electrical).
- iii) In case of different agencies having taken work permit for the same equipment e.g. Air Fin Cooler – motor by Electrical Maintenance, and Fan Blades by Mechanical Maintenance; UIC shall ensure that work permits from all the agencies involved are returned back, before energising the equipment.

Each such agency shall be issued separate work permit along with lock out and tag out (LOTO) on respective equipment. Unless all the work permits are returned back to UIC, locks & tags are removed, the equipment shall not be energised.

- iv) Before issuing any permit for single equipment having more than the one source of electric supply, UIC shall arrange through Electrical-in-charge for:
 - a. Positive Electrical Isolation (by disconnection of all sources of electric power supply) and obtaining Electrical Isolation Line Clearance. All the respective outgoing at the source panel themselves are locked and tagged properly.
 - b. Effective earthing of line/ equipment (Isolated section).
- v) After completion of the work, EIC shall arrange to deenergise and remove all temporary installations, construction materials, debris and withdraw personnel from site, and returns Work Permit/Clearances to UIC. UIC shall arrange for safe charging of the equipment after obtaining Electrical Energisation Line Clearances.
- vi) Throughout the work, designated person shall ensure that the switches and circuit breakers controlling the supply to the equipment or line, shall be kept tagged with caution boards stating **MEN WORKING ON EQUIPMENT** to caution against anyone inadvertently closing them. The control switches/ breakers, wherever possible, should be locked and in racked out position. The area of work shall be strictly cordoned off from unauthorised entry.

	OISD – STD – 137	Page No. 6
	INSPECTION OF ELECTRICAL EQUIPMENT	

- vii) While carrying out Electrical Isolation/ Energisation work, the persons connected with the work shall use only approved and tested protective equipment (such as Rubber gloves, rubber shoes, safety belts, ladders, earthing devices, helmets, etc.,) for protection against mechanical and electrical injury, taking all the safety precautions described in IS:5216. Such Tools and devices shall always be maintained in sound and efficient working condition. Provision of the Insulating Floor or mats conforming to IS-15652:2006 is to be adhered.
- viii) Electrical Line Clearances shall be taken even while working in proximity of bare energized electrical conductor.
- ix) Each line clearance Energisation permit Certificate shall be used for one equipment only.
- x) For non-routine jobs, Job safety Analysis (JSA) should be carried out. JSA should involve all the working personnel including non-electrical persons (but who requires electrical isolation for their job), Designated Electrical Person, UIC, Electrical In Charge and EIC. The program, sequence, steps & tasks of the job, various hazards involved in each task and their mitigation measures shall be discussed and recorded. This record can be attached to the work permit. The same shall be followed strictly during the execution of the job, considering the scenario changes at site at that time of execution.
- xi) Where ever the source of supply is fed from long distance and it is practically not possible to get the Electrical Line clearance / Energising Certificate in person, then telephonic clearance/ recording Electrical Lockout/ Message No. at both ends can be adopted. Owner organisation shall document and establish such a procedure and are to be followed as per IS 5216.
- xii) Overriding of any safety related interlocks for the purpose of testing even under isolated condition shall be carried out under the special written permission from appropriate authorities. The owner organisation shall have documented and established procedure, approved by competent authority. A record shall be maintained to track such overrides. Once the testing job is over, safety interlocks shall be normalised immediately.

4.6 PROCEDURE FOR EARTH EXCAVATION CLEARANCE

This Clearance shall be taken by EIC from concerned departments like electrical, communication, civil, process, fire & safety etc.), wherever earth excavation is to be carried out. Without such clearance, if excavation is carried out, especially over underground facilities (Cables, Pipelines etc.) it may cause fatal accidents to the personnel, besides adversely affecting the plant operation due to possible damage of such underground facilities (e.g. piercing the insulation of live cable through sharp tool like pickaxe/ heavy excavation equipment).

- i) Based on clearances from the concerned departments, UIC will issue the work permit to EIC in an approved format to be developed by individual organisations depending on the type of installations and administrative set up. While designing the format, the need of owner-in-charge concept shall also be taken care of, as defined in OISD-STD-105 (Work Permit System).
- ii) UIC/ Respective departments shall clearly spell out the precautions to be taken, while carrying out such earth excavation, in the Clearance certificate itself. Prior to issue of Clearance certificate, he shall ensure that the proposed excavation does not interfere with existing underground facilities such as power (H.T & L.T),

	OISD – STD – 137	Page No. 7
	INSPECTION OF ELECTRICAL EQUIPMENT	

telephone, fire alarm, lighting, and control cables, earthing, and cathodic protection installations, OFCs, process pipelines, etc.

- iii) EIC shall ensure compliance of all the safety measures including those stipulated in the Clearance and provide supervision throughout the work for safe excavation and protection of the underground facilities from any damage due to excavation. He shall promptly report to UIC/ concerned departments for any damage to underground facilities even if it be of minor nature, and shall proceed with the excavation work only after complying with further instructions, as endorsed in Permit, if any.
- iv) UIC shall ensure that all the clearances obtained by EIC have been returned and facilities are normalised before closing the Work Permit.

5.0 BASIS AND TYPES OF INSPECTION

5.1 BASIS OF INSPECTION

5.1A Failures of electrical equipment gives rise to any or all the following hazards:

- i) Fatal accidents due to electric shock.
- ii) Fire accidents and burn injuries due to electric arcs/sparks/hot surfaces
- iii) Abrupt interruption in plant operation.
- iv) Environmental hazards such as leakage of insulating oil / SF6 gas etc.
- v) Burn injuries due to leakage of hot lube oil/ transformer oil.
- vi) Injuries to the personnel due to due to improper handling of charged springs of various circuit breakers.
- vii) Fire/ spark/ burn/ explosions due to wrong operations of battery bank and capacitor bank.
- viii) Flashover due to failure of shrouds between phases in a cubicle.

5.1B The failures of electrical equipment can be attributed to:

- i) Imposed external conditions such as deterioration of enclosure, insulation and operating mechanism by environment conditions like heat, moisture, dust, chemical attack, dirt, drop object, cable damage due to vehicle/ crane entry etc.
- ii) Quality of power supply (like undervoltage, overvoltage, frequency variation, asynchronous operation, improper wave form e.g. main supply to inverter supply, overheating, power dips, unbalanced voltage, frequently failures of power supply etc.)
- iii) Overloading due to any reason.
- iv) Internal faults in the equipment itself mainly due to manufacturing defects, or those crept in during installation, maintenance and repairs.
- v) Inferior quality of work due to factors like poor workmanship by unqualified/unskilled and untrained workers, inadequate or incompetent

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	OISD – STD – 137	Page No. 8
	INSPECTION OF ELECTRICAL EQUIPMENT	

supervision, non-compliance of safety and statutory requirements, adoption of incorrect procedures, and usage of poor quality of tools and tackles.

- vi) Usage of substandard materials and equipment not in conformity with original design specifications.
- vii) Design deficiencies in the technical specifications, improper sizing and layout of installations and lack of total system concept.
- viii) Bypassing of protective and interlocking devices/circuits during operation though originally designed for the system.
- ix) Inability for carrying out shutdown inspection/tests on the equipment in time, as required by maintenance inspection schedules, due to difficulties in taking the equipment out of service.

Inspection programmed on electrical equipment and associated circuits shall be devised on the basis that the inspection results will be able to pinpoint the trouble spots in the installations, well in advance, allowing ample time for the maintenance personnel to take remedial measures, so as to prevent failure of equipment leading to unsafe conditions and hazards as mentioned above. No equipment shall be operated beyond its safe life period assessed on the basis of inspections from time to time and optimum replacement schedule shall be followed. Manufacturer's recommendation should also be reviewed, wherever available, before replacement decisions are made.

5.2 TYPES OF INSPECTION

The field inspection of electrical equipment in an operating unit is classified into following three types:

i) ONSTREAM INSPECTION

This covers audio visual inspection items (instrument aided or otherwise) for checking of general equipment conditions, while the equipment is in operation. (e.g. an abnormal noise or vibration of motor, leakage of transformer oil or overheating of equipment etc., can be assessed through such inspections).

ii) SHUTDOWN INSPECTION

This covers those inspection items for checking conditions of equipment and systems which cannot be revealed through onstream inspection (such as internal condition of equipment). This inspection shall be carried out after taking the shutdown of the related equipment after obtaining the Electrical Line Clearance, and Work Permit, requirements of which are detailed in Section 4.0 and in OISD-STD-105 (Work Permit System)

iii) INSPECTION OF STANDBY EQUIPMENT AND SPARE PARTS

The standby electrical equipment shall be inspected periodically. A scheduled change over concept from running equipment to standby equipment shall be in place to ensure the equipment's technical and process integrity. Preferably standby rotary electrical equipment should not remain idle for more than 2 months. The standby rotary equipment remaining idle for more than 6 months, should additionally be checked for-

- a) Free rotation of shaft
- b) Proper functioning of space heater circuit
- c) Weather protection.
- d) Insulation resistance of motor with cable.

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	OISD – STD – 137	Page No. 9
	INSPECTION OF ELECTRICAL EQUIPMENT	

If, however, standby equipment was not energised earlier, all applicable Precommissioning checks, as detailed in Section 9.0 shall be carried out before taking it into service.

The spare parts shall be inspected to ensure compliance with original specifications as per manufacturers' catalogue number.

6.0 FREQUENCY OF INSPECTION

The inspection scheme given in the following tabular format shall be followed as minimum requirement for the items to be inspected. Frequencies of inspection in this format have been arrived at, considering a typical installation and therefore shall be treated as guidelines. Depending on specific requirements of the installation, considering severity of usage, criticality of equipment, environmental conditions and such other factors, individual organisations shall review and develop own inspection frequency standards which shall be strictly followed.

The frequency of inspection for new or special equipment, where previous experience is not available, shall be reviewed by the concerned organisation, after one or two inspections and reconsidered whether the interval could be lengthened safely or should be shortened. Care shall be taken against the bad effects of over maintenance especially for flameproof and such critical installations. No attempt should be made to replace or repair a glass in a flameproof enclosure (e.g. in lighting fixture or the parts) except with the complete assembly or part obtainable from the manufacturer for which the flameproof certificate from the competent authority shall be obtained to confirm its flame proofness.

For inspection procedures, Section 7.0 shall be referred. The safety Electrical Line Clearances as required in Section 4.0 shall be obtained for work where Electrical Isolation is needed or when any work is to be carried out in close proximity to bare energised electrical circuits.

Insulation resistance, earth resistance and continuity tests, when required to be carried out in a hazardous area, shall be carried out only with certified intrinsically safe insulation/earth tester suitable for use in classified hazardous area, after taking the shutdown of the related equipment through permit system.

However, if such hazardous area is rendered non hazardous and certified by competent authority to be a safe area, usage of standard testing equipment shall be permissible till such time the area continues to remain non-hazardous.

Additionally, the applicable provisions in the API Guide Chapter XIV (electrical systems) Table 5 to 13 (methods of Inspection) and the recommendations of the manufacturer shall be followed, while devising inspection schedules.

LEGEND:

D : Daily
W : Weekly
F : Fortnightly
M : Monthly
Q: Quarterly

HY : Half yearly
nY : Once in n years
TA : Turn around
ASCO: After specified cycles of operation
BU: Before use

Wherever, the monitoring of the system is through SCADA, it should be ensured that monitoring of critical data is maximised through on-line data acquisition system.

	OISD – STD – 137	Page No. 10
	INSPECTION OF ELECTRICAL EQUIPMENT	

6.1 ROTATING MACHINERY

Items to be inspected	Frequency
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6.1.1 Generator (including Exciter)

6.1.1A The following shall be Onstream Inspection items:

i)	Condition of commutator (for sparking), brushes, and collector ring). Due care to be taken for hydrocarbon gas presence, hot spot, and inspection of spark on commutator with naked eye.	W
ii)	Bearing condition (lubrication, noise, and temperature)	W
iii)	Cooling system condition.	W
iv)	Housekeeping	W
v)	External damages	W
vi)	Leakage of Oil/Water	F
vii)	Surface overheating	F
viii)	Ventilation	F
ix)	Abnormal noise	F
x)	Functioning of local meters & gauges	F
xi)	Corrosion effects (external)	HY
xii)	Provision of Guards to moving parts	M
xiii)	any other inspection recommended by the manufacturer	

Note:

For critical generators such as for Captive power plants, the frequency of inspection shall be Daily (D) basis for all items, except for items (XIII).

6.1.1B The following shall be Shutdown Inspection items (during scheduled maintenance):

i)	Insulation resistance
ii)	Earthing connections (checks shall include earth resistance measurements, tightness of connections, and continuity of conductors)
iii)	DC resistance of stator/ rotor / exciter**
iv)	Air gap between stator and rotor**
v)	Manual and automatic synchronising and change-over systems.
vi)	Alignment of coupled machine
vii)	Operation of circuits especially related to space heater, temperature detector, tripping devices, annunciators and interlocks.
viii)	Availability of all protective devices, and circuits in line as per design, and all in operating condition
ix)	Moisture absorption (Dielectric Absorption Test)*
x)	Insulation deterioration (Polarisation Index Test)*
xi)	Internal damages, discolouration, open or short circuits, and loose connections**
xiv)	Condition of commutators, brushes, and collector rings
xv)	Condition of foundation and tightness of foundation bolts
xvi)	Any other inspection recommended by the manufacturer

Note :

* Items (ix) and (x) are needed for critical high voltage equipment only.

** items to be checked during major overhauling

	OISD – STD – 137	Page No. 11
	INSPECTION OF ELECTRICAL EQUIPMENT	

6.1.2 Motor (H.T and L.T)

6.1.2 A The following shall be Onstream Inspection items:

i)	General Cleanliness	M
ii)	Corrosion effect (external)	HY
iii)	Surface Overheating/sparking	M
iv)	Ventilation	M
v)	Vibration	M
vi)	Abnormal noise	M
vii)	External damages	M
viii)	Load (Amperes)	M
ix)	Lubrication	Q
x)	Cooling System	Y
xi)	Provision of Guards to moving parts	M
xii)	Condition of commutator, brushes, and collector rings for slipring and synchronous motors	M
xiii)	Healthiness of purging system for Ex 'p'	M
xiv)	Effect of supply system disturbances such as voltage drop, power dip etc. (as given in the log entries at substation)	
xv)	any other inspection recommended by the manufacturer	

Note:

For critical motors, the frequency of inspection shall be weekly (W) basis in case of all items (i) to (xv).

6.1.2B The following shall be Shutdown Inspection items (during scheduled maintenance):

i)	Mechanical freeness of rotor shaft
ii)	Insulation resistance
iii)	Winding resistance (Measurement)
iv)	Earth resistance and continuity
v)	Operation of -- Space heater -- Temperature detector -- Interlocks/trips -- Annunciation circuits
vi)	Availability of all protective devices and circuits in line as per design, and all in operating condition.
vii)	Condition of remote tripping devices & limit switches
viii)	Limit & torque switches (in the case of motor operated valves) for proper functioning.
ix)	Oil level in lubricating gear case.
x)	Moisture absorption (Di-electric Absorption Test)*
xi)	Insulation deterioration (Polarisation Index Test)
xii)	Internal damage, discolouration and loose connection**
xiii)	Stator to rotor air gap measurement**
xiv)	Condition of commutator, carbon brushes(for correct brush pressure) and collector rings
xv)	Condition of supporting legs, pedestal and base plate
xvi)	Condition of the foundation and tightness of foundation bolts.
xvii)	Any carryover of oil or grease inside the windings
xviii)	Signature analysis (optional)
xix)	Tan delta test (optional)

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	OISD – STD – 137	Page No. 12
	INSPECTION OF ELECTRICAL EQUIPMENT	

xx) Any other inspection recommended by the manufacturer

Note:

*Items (x) are needed for critical high voltage equipment only.

** items to be checked during major overhauling

In the case of Synchronous/Slipring motors, following additional Shutdown Inspection items shall be carried out during major overhauling:

i)	Tightness of the DC Poles fitted on the outer-Periphery of the rotor
ii)	Condition of end rings
iii)	Continuity of the leads connecting the poles and the slip rings via brushes
iv)	Any other inspection recommended by the manufacturer

6.2 STATIONARY EQUIPMENT

Items to be inspected	Frequency
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6.2.1 Switchboard and Panel

6.2.1A The following shall be Onstream Inspection items:

i)	General Cleanliness	W
ii)	Weather protection	W
iii)	Dust and vermin-proofness	W
iv)	External damages	W
v)	Control supply availability	W
vi)	Ventilation system	W
vii)	Corrosion effect (external)	W
viii)	Accessibility for maintenance /operation	M
ix)	Functioning of meters, and lamp indicators	M
x)	Oil level and leakage	M
xi)	Correctness of relay settings (for critical services)	TA
xii)	Thermography of panels, as applicable	HY
xiii)	Gas pressure of the enclosure for GIS switchgears	W
xiv)	Insulating Mats checking	Y
xiii)	Any other inspection recommended by the manufacture	

6.2.1B The following shall be Shutdown Inspection items (during scheduled maintenance):

i)	Dielectric strength of oil (in breakers) and Oil Contamination
ii)	Condition of earthing connection including earthing between panel door and panel.
iii)	Insulation and earth resistance
iv)	Electrical resistance of breaker
v)	Internal damages (breakage of barriers, arc chute cracks, jamming solenoid, dirt/dust accumulation in any part etc.)
vi)	Operation of -- Protection, control and interlock circuits -- Circuit breaker (rack-in-out mechanism, springs of driver) -- Simultaneous closing/opening of all the

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	OISD – STD – 137	Page No. 13
	INSPECTION OF ELECTRICAL EQUIPMENT	

	three phases. -- Contactor operation -- Space heater operation -- Safety shutters -- Adjustment of limit switches
vii)	Tightness of all bolts, cable connections, and terminals
viii)	Condition of contacts, contact alignment, and clearances with live parts
ix)	Breaker closing and opening time & total relay operating time
x)	Checking and calibration of relays
xi)	Checking of VCB bottles for healthiness (Inspection frequency 3Y)
xii)	For GIS switchgears, the inspection items shall be as per OEM recommendations.
xi)	Any other inspection recommended by the manufacturer

6.2.2 Transformer, Reactors & Bus duct

6.2.2A The following shall be Onstream Inspection items:

i)	Oil leakage	W
ii)	Oil level (in transformer, bushing and conservator tank)	W
iii)	Condition of silica gel	W
iv)	Oil/ Winding temperature meters working	W
v)	Load (Amperes)	W
vi)	Cleanliness	W
vii)	External damages	HY
viii)	Operation of cooling fans & surface temperature	W
ix)	Annunciation (for abnormal conditions)	TA
x)	Condition of cable end sealing and terminal boxes	Y
xi)	Insulators and bushing (check for any cracks) for outdoor transformers	TA
xii)	Corrosion effect (external)	HY
xiii)	Condition of oil soak pit and oil drainage arrangement	Y
xiv)	Sealing of bus duct entry on both sides of wall (for vapour /dust/water tightness)	Y
xv)	Visual inspection of transformer foundation and yard up keeping.	Q
xvi)	Transformer oil testing as per IS:1866	Y
xvii)	OLTC oil testing as per IS:1866	Y
xviii)	DGA of Transformer oil	3Y
xix)	Checking of water sprinkler/ NIFP for transformer fire fighting system.	Y
xx)	Any other inspection recommended by the manufacturer	

6.2.2B The following shall be Shutdown Inspection items (during scheduled maintenance):

i)	Terminal connection in primary & secondary
ii)	Operation of protection and annunciation circuits
iii)	Internal damages, condition of supports, bus duct, flexibles and insulators
iv)	Insulation resistance of primary and secondary windings
v)	Earth resistance and earth continuity
vi)	Condition of Explosion vent diaphragm
vii)	WTI & OTI calibration
viii)	Transformers Cooling fans auto start interlock testing
ix)	Checking of Buchholz relay operation
x)	Completeness with all accessories

	OISD – STD – 137	Page No. 14
	INSPECTION OF ELECTRICAL EQUIPMENT	

xi)	NGR checking
xii)	OLTC M&I as per OEM recommendation
xiii)	Tan-delta testing of Transformer/ Reactor – 3Y
xiv)	Any other inspection recommended by the manufacturer

6.2.3 Battery

6.2.3A The following shall be Onstream Inspection items:

i)	General cleanliness of battery and battery room	W
ii)	Functioning of ventilation system	W
iii)	Spillage of electrolyte	W
iv)	Level of electrolyte	M
v)	Cell voltage	M
vi)	Specific gravity of cell (for lead acid batteries)	M
vii)	Availability & condition of safety equipment (such as rubber apron, goggle, and acid proof hand gloves)	M
viii)	Condition of acid-proof flooring	M
ix)	No electrical fitting on top of Battery Bank (Electrical fittings shall be on the wall)	M
x)	Battery load test (minimum curative discharge@C10 for 2 hrs)	Y/as per OEM
xi)	Availability of Tap water in Battery room.	M
xii)	Condition of Battery Stand	M
xiii)	Any other inspection recommended by the manufacturer	

Note:

For critical/emergency and switchgear power plant applications, the frequency of inspection shall be Daily (D) basis, in case of items (i) to (v) and weekly (W) basis, in case of items (vi) to (xiii)

6.2.3B The following shall be Shutdown Inspection items:

i)	Joints, terminal and bolted connections for tightness (for full electrical contact)
ii)	Blockage of vent plugs
iii)	Corrosion effects on terminals and connectors
iv)	Condition of supports (racks)
v)	Damages such as cracks in containers
vi)	Capacity test
vii)	Any other inspection recommended by the manufacturer

6.2.4 Battery Charger & D.C. Board

i)	General cleanliness	W
ii)	Ventilation system	W
iii)	Float/Boost voltage & charging current	W
iv)	Overheating	W
v)	External damages	W
vi)	Tightness of connections	TA
vii)	Condition of contactors and switches	TA
viii)	Operation of charger circuit with annunciation	TA
ix)	Changeover circuits (emergency backup)	TA

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	OISD – STD – 137	Page No. 15
	INSPECTION OF ELECTRICAL EQUIPMENT	

x)	Earth leakage monitoring in Battery Charger	D
xi)	Any other inspection recommended by the manufacturer	

Notes:

- For critical/emergency switchgear, and power plant applications the frequency of inspection shall be Daily (D) basis for items (i) to (v)

6.2.5. VFD/ U.P.S/Invertor

6.2.5A The following shall be Onstream Inspection items:

i)	General cleanliness, (accumulation of dust, and dirt on contactors, diodes, thyristors, transformers and other components) and external damage	W
ii)	Condition of ventilation and detection of any overheating	W
iii)	Dust free and temperature controlled environment in UPS/VFD/Inverter room (Indoor Installations)	W
iv)	Water level checking in VFD (for water cooled VFDs)	W
v)	Thermography of panels, as applicable	HY
vi)	Preventive maintenance and condition monitoring to be done as per the schedule specified by OEM	TA
vii)	Earthing connection healthiness including earthing between panel door and panel.	TA
viii)	Checking of cooling system and its flow	W
ix)	Cooling water parameters of VFD	W
x)	Any other inspection recommended by the manufacturer	

6.2.5B The following shall be Shutdown Inspection items (during scheduled maintenance):

i)	Internal damages including condition of heat exchangers.
ii)	Pitting of contacts
iii)	Voltage and wave forms as specified by the manufacturer
iv)	Tightness of wiring connections
v)	Operation of protective devices and changeover circuits
vi)	Functional checks for all circuits, and annunciation
vii)	Any other inspection recommended by the manufacturer

6.2.6 Capacitor Bank

6.2.6A The following shall be Onstream Inspection items:

i)	General cleanliness	Q
ii)	Structural support	Q
iii)	Damages (external)	Q
iv)	Condition of ventilation	Q
v)	Oil level	Q
vi)	Oil leakage	Q
vii)	Grounding connections	Q
viii)	Thermography	HY
ix)	Any other inspection recommended by the manufacturer	

	OISD – STD – 137	Page No. 16
	INSPECTION OF ELECTRICAL EQUIPMENT	

6.2.6B The following shall be Shutdown Inspection items:

i)	Insulation resistance
ii)	Discolouration on cable connector clamps
iii)	Operation of protective device
iv)	Tightness of wiring connections
v)	Corrosion effects
vi)	Any other inspection recommended by the manufacturer

6.2.7 Switchyard Equipment (Circuit breaker, Isolator, C.T, P.T, L.A., Support Insulators)

6.2.7A The following shall be Onstream Inspection items:

i)	Overheating, and sparking of contacts	W
ii)	Oil level & leakage/SF6 availability	W
iii)	Visual inspection of Earthing connections, and overhead earth mesh condition	W
iv)	General Cleanliness	W
v)	Equipment Damages	W
vi)	Corrosion effects (External)	HY
vii)	Functioning of lamps and meters	M
viii)	No dry grass in switchyard	W
ix)	Any other inspection recommended by the manufacturer	

6.2.7B The following shall be Shutdown Inspection (atleast once in a year) items:

i)	Operation of switchyard equipment with control, trip, and annunciation circuits
ii)	Condition of insulators
iii)	Alignment between contacts for Isolators
iv)	Insulation resistance
v)	Earth resistance
vi)	Any other inspection recommend by the manufacturer

6.2.8 Lighting Fixture & Junction-box (installed for plant/yard/street and building lighting.)

6.2.8A The following shall be Onstream Inspection items;

i)	Cleanliness & external damages	W
ii)	Fixture support	W
iii)	Lighting Illumination level	Y
iv)	Condition of conduits, Junction boxes	M
v)	Plugging of un-used cable entries	HY
vi)	Changeover circuits (Emergency lighting)	M
vii)	Any other inspection recommend by the manufacturer	

6.2.8B The following shall be Shutdown Inspection items:

i)	Condition of gaskets including cleanliness
ii)	Operation switches & circuits
iii)	Insulation resistance
iv)	Earthing connections

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	OISD – STD – 137	Page No. 17
	INSPECTION OF ELECTRICAL EQUIPMENT	

v)	Loose connections
vi)	Corrosion effect, condition of paint (lighting fixtures, junction boxes, conduits, poles and switches)
ix)	Any other inspection recommended by the manufacturer

6.2.9 Test Instruments for Inspection

The following inspection items (common for all the instruments listed) shall be carried out:

- Cleanliness
- Breakages (of glass cover etc.)
- Calibration test, and other safety checks before use as per manufacturer's recommendation
- Zero setting, errors
- Bent pointers, sticky or jerky movement (Analog meters)
- Instrument Battery condition

List of instruments (with the frequency of inspection marked against each item) is given below:

i)	H.V. discharge rod	M/BU
ii)	Non-Contact Voltage detector	M/BU
iii)	Insulation Tester	M
iv)	Earth Resistance tester	M
v)	Cable fault locator	M
vi)	Multimeter	M
vii)	Tong tester	M
viii)	Relay test bench	Y
ix)	Motor test bench	Y
x)	Oil filtering equipment	Y/BU
xi)	Dielectric test equipment	Y/BU
xii)	Phase sequence meter	M/BU
xiii)	Portable voltmeter, ammeter, frequency meter, hydrometer, thermometer	M/BU
xiv)	H.V. testing unit	Y/BU
xv)	Lux meter	Y
xvi)	Infrared gun for Thermography	Y
xvii)	Oscilloscope	Y
xviii)	Non-Contact Temperature scanner	Y
xix)	RPM meter	Y

6.3 PROTECTIVE SYSTEMS

Items to be inspected	Frequency
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6.3.1 Relay & Meter

6.3.1A The following shall be In-service Inspection items:

i)	Cleanliness	W
ii)	Breakages of glass & cover	M

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	OISD – STD – 137	Page No. 18
	INSPECTION OF ELECTRICAL EQUIPMENT	

iii)	Erratic operation (observation from log book entries)	D
iv)	Any other inspection recommended by the manufacturer	

6.3.1B The following shall be Shutdown Inspection items:

i)	Trip circuit operation (simulation test)	TA
ii)	Relay operation under fault conditions simulated (primary injection)	TA
iii)	Correctness of relay settings	TA
iv)	Changeover circuit operation	TA
v)	Checking and calibration of all meters (KWA/KVA demand, ammeter, volt meter)	Y
v)	Electromechanical relays testing	Y
vi)	Numerical relays testing	5Y
vii)	Any other inspection recommended by the Manufacturer.	

6.3.2 Earthing & Bonding

6.3.2A The following shall be General Inspection items:

i)	Visual checks for external damages to earth conductors, earth electrodes and earthing connections.	HY
ii)	Operation of earth leakage annunciation and trip circuits for lighting, power circuits and appliances (Simulation Tests)	Y
iii)	Corrosion effects and tightness of connections	HY
iv)	Identification of earth pits in the field with permanent markers.	HY
v)	Earth resistance on individual electrodes once in dry and one in wet weather	HY
vi)	Continuity of earthing conductors	Y
vii)	Bonding and earthing of all metallic structures, pipe racks, armour of cables, and electrical equipment enclosures	Y
viii)	Earth loop impedance (measurement)	TA

6.3.2B The following shall be the Special items for Inspection, applicable to different categories of installation:

i)	Neutral earthing of the electrical system	Y
ii)	Body earthing of the electrical equipment	Y
iii)	Tankage (bonding of external floating roof, ladder rollers, bypass conductors, external shunts, rails of staircase and lubrication on rollers)	HY
iv)	LPG bottling plants, loading gantries /tank truck loading Check for: a) Earthing continuity of loading arms, flexible hoses, and piping b) Earthing & bonding of rail tracks within the gantry premises c) Earth connections in carousal /tanks	HY
v)	Process equipment in the plant area (such as heat exchanger, vessel, column, piping etc.) whether grounded	Y

	OISD – STD – 137	Page No. 19
	INSPECTION OF ELECTRICAL EQUIPMENT	

6.3.3 Lightning Protection

i)	Visual check for external damages to earth electrodes down conductors, lightning arrestors, supporting structures and insulators.	Y
ii)	Cleanliness of insulators	Y
iii)	Earth resistance and continuity	Y
iv)	Any other inspection items required as per IS:2309 or BS 6651 as applicable	Y
v)	Third harmonic current testing of lightning arrestor	3Y

6.3.4 Flameproof/ Weather protection

i)	Visual checks for external damages (to glasses, casting etc.)	M
ii)	Visual Inspection of external earthing connections	Q
iii)	Condition of gasket	Q
iv)	Checking unauthorised modifications in the flameproof equipment.	Y
v)	Checking completeness of bolts, glands, stoppers & their tightness and the enclosures having intended weather protection	Y
vi)	Corrosive effects	Y
vii)	Plugging spare cable entries	Q
viii)	Any other inspection as recommended by the Manufacturer or system designer.	

6.3.5 Fire alarm system (Electrical circuits)

i)	Operation of siren circuits (simulation test)	W
ii)	Operation of break glass field station actuation circuits	W
iii)	Checks to confirm no blown fuses, and no fused lamps	M
iv)	Checking of emergency supply system availability	W
v)	Any other inspection recommended by the manufacturer or system designer	

6.3.6 Cathodic protection (Electrical circuits)

For tankages, underground piping, submerged structures and cross-country pipelines)

i)	Performance of transformer-rectifier unit (TRU) and earthing connections	W
ii)	Pipe/structure to soil potential measurements for impressed current / sacrificial anode systems	HY
iii)	Anode current and all connections to anode / structures (conformity to design requirements)	HY
iv)	Any other inspection recommended by the Manufacturer or system designer	

Note:

The frequency of inspection is to be increased if abnormal external corrosion and breakdowns are encountered in the cathodically protected system.

	OISD – STD – 137	Page No. 20
	INSPECTION OF ELECTRICAL EQUIPMENT	

6.3.7 Electric Heat Tracing.

i)	Visual checks for functioning of all panel indication lamps (indicating tracer circuit healthy condition)	W
ii)	Temperature measurements at inspection point	W
iii)	Current reading of each circuit and check for any blown out fuse	W
iv)	External damages to piping, insulation and tracers	M
v)	Weather proofness for outdoor panels and junction boxes especially whether doors / covers kept closed.	M
vi)	Tightness of connections*	
vii)	Sealing and termination*	HY
viii)	Insulation resistance of cables, panel boards and tracers*	
ix)	Continuity checks*	
x)	Any other inspection recommended by the manufacturer or system designer.	

Note:

*Shutdown Inspection items

6.3.8. Protective and Safety Appliances

The condition of following appliances shall be checked as per the requirements of IS: 4770 or as recommended by the manufacturer. The frequency of inspection shall be 'HY' or everytime before being put to use, whichever is earlier.

- I) Hand Gloves, Safety suits, breathing apparatus etc.
- II) Rubber shoes, insulating mats
- III) Safety belts
- iv) Ladders
- v) Earthing devices
- vi) Helmets
- vii) Voltage detectors/ Multimeters/ Tong testers/ Insulation tester/Earth resistance tester etc.
- viii) Goggles & face shields
- ix) Earth leakage circuit breakers (ELCB)/ Residual current circuit breakers (RCCB).

6.4 DISTRIBUTION NETWORK

Items to be inspected	Frequency
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6.4.1. Cable and conduit wiring

i)	Availability of permanent cable route markers and joint location markers in the field	Y
ii)	Condition of cable termination /joints and clamping arrangements	Y
iii)	Availability of identification tags	Y
iv)	Mechanical protection to under ground and overhead cables	Y
v)	Damages to cables, conduits, wires including clamps/cable supports	TA
vi)	Water tightness of conduit system	TA
vii)	Corrosive effects on conduits & junction boxes.	TA

Notes:

- It shall be checked (through random trial pits) that the underground cables are protected with sand and bricks throughout and ensure that the cables are not

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	OISD – STD – 137	Page No. 21
	INSPECTION OF ELECTRICAL EQUIPMENT	

exposed at any location. Also Armouring, pipe sleeves and such other mechanical protection requirement shall be inspected.

2. Item vi) shall be applicable only for critical high voltage cables.
3. Items (v) to (vii) shall be Shutdown Inspection items.

6.4.2 Cable Duct/Trench/Tray support

i)	Congestion of cable in the duct (conformity for pipe fill 50%) – during initial installation and laying of new cables.	
ii)	Damages to cable trench including covers, duct, tray support and walkway gratings.	Y
iii)	Sealing cable trenches /ducts (prevention of entry of foreign materials-vermin, water, chemicals, dirt, reptiles, etc into trenches, ducts)	Y
iv)	Corrosion on the metallic tray supports	Y

6.4.3 Overhead line (Uninsulated)

i)	Cleanliness of insulators and bushings	Q
ii)	Condition of hardware, supporting structure and conductors	Q
iii)	Abnormal sagging and clearances	Q
iv)	Joints in the O.H. lines	Q
v)	Continuity of earth conductors	Q
vi)	Provision of guards, anticlimbing devices, danger boards, and phase markings	Q
vii)	Corrosion effects	Q
viii)	Infringement and growth of vegetation	Q
ix)	Any damages to insulators and other devices	Q
x)	Earthing of stay wire	Q
xi)	Physical inspection of Overhead line conductors (of both normal and standby O.H. lines) including ground clearance of guard wire.	TA

6.5 OUTDOOR SWITCHBOARD

Items to be inspected		Frequency
i)	Earthing of fencing, gage, insulator supports and other metallic structures	HY
ii)	Condition of fencing & all metallic structures from corrosion effects	HY
iii)	Checking of pipe from transformer to soakpit for any leakage	HY
iv)	Provision of locking arrangement for gate and isolators (especially when the transformers & O.H. line are energised)	HY
v)	Provision of metal mesh, oil soakpit and fire partition walls (as applicable)	HY
vi)	Condition of trenches cover slabs and earth electrodes in the yard	HY
vii)	Visual inspection of transformer plinth and other concrete foundations	TA
viii)	Accessibility for maintenance and operation of switchyard equipment with mountings.	HY
ix)	Illumination level	Y

	OISD – STD – 137	Page No. 22
	INSPECTION OF ELECTRICAL EQUIPMENT	

6.6 SWITCHGEAR ROOM/SUBSTATION BUILDING

Items to be inspected		Frequency
i)	Illumination level	HY
ii)	Emergency lighting provision and its auto-changeover circuit operation	HY
iii)	Ventilation and Pressurisation system.	HY
iv)	Roof leakage - Pre-monsoon Pond test	Y
v)	Sealing of cable and bus duct entries (to make them water and vapour tight)	Y
vi)	Housekeeping including movement of door/ shutters.	M
vii)	Water drainage arrangement	Y
viii)	Provision of insulating mats conforming to IS:15652:2006 having ISI mark, shock treatment chart, fire extinguisher (suitable for electrical fires), updated SLD, sand filled buckets, MCP, Earthing rods, Electrical PPEs, and telephone	HY
ix)	Condition of trench, cover slabs, and chequered plates	HY
x)	Marking on the switchboard for identification of circuits and provision of danger, caution board. Marking should be across the fixed and removable portion of the rear side of the panel for safety.	HY
xi)	Damages and corrosion effects, if any	HY

6.7 SAFE PROCEDURES AND SYSTEMS

Checks for compliance:

i)	Compliance of Work Permit System	HY
ii)	Safety procedures adopted by external agencies	HY
iii)	Unauthorised additions or modifications of installations (whether temporary or permanent)	HY
iv)	Adherence to safety and fire fighting requirements as per standards	Y
v)	Compliance of items pointed out by Electrical Inspector during periodical statutory inspection (CEA Regulation 30)	Y
vi)	Physical inspection of electrical equipment installed in hazardous areas for integrity (tightness of enclosure nuts& bolts, sealing of un-used cable entries and earthing).	Y
vii)	Availability of updated operating and maintenance manuals	Y
viii)	Safety audit systems (internal & external)	Y
ix)	Training programs of electrical staff.	Y
x)	Availability of adequate illumination	Y
xi)	Preservation of idle equipment, cables, and spare parts	Y
xii)	Procedure for allowing temporary electrical facilities	Y
xiii)	Adherence to logging system of failures, system disturbances, break-downs, trippings and corrective actions taken.	Y
xiv)	Accessibility for operation and maintenance of electrical facilities	Y

7.0 PROCEDURES FOR ELECTRICAL TESTS AND RECOMMENDED VALUES

This section provides brief details of the tests normally carried out in field to determine the condition of the various electrical equipment/ system. The tests provide data to determine whether any corrective maintenance or replacement is necessary or desirable, to ascertain

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	OISD – STD – 137	Page No. 23
	INSPECTION OF ELECTRICAL EQUIPMENT	

the ability of the element to continue to perform its design function adequately and finally to chart the gradual deterioration of the equipment over its service life.

The test results may vary depending on the age, site, installation & maintenance practiced followed. The results shall be judiciously used to enhance life of the equipment and systematic plan for its replacement.

Before initiating the inspection of electrical equipment and associated circuits, concerned engineer should familiarise himself with the complete previous history of the equipment/circuits, design parameters, service, likely areas of concern, manufacturer's recommendations and such other pertinent data to arrive at the appropriate inspection procedures/ program.

Many tests on the electrical equipment involve the use of high voltages and currents that are dangerous; hence safety rules should be instituted and practiced to prevent injuries to personnel performing the test. Also, the test procedures used should be designed to ensure that no intentional damage to equipment will result from the testing process.

Most of the defects/faults in electrical system can be discovered only through tests and not by external inspection alone.

For the basis on the type of inspection, Section 5.0 shall be referred. For competence of the person conducting the tests, Section 3-ii) shall be referred. Unless otherwise mentioned, the acceptable test values shall be the compliance of CEA Regulation 12 and the related standards published by Bureau of Indian Standards.

I) INSULATION RESISTANCE TEST:

Insulation tests are used to determine the quality or condition of the insulation systems of electrical equipment or circuits.

Insulation resistance is measured to ensure absence of leakage current and humidity. For a given insulation condition, the acceptable value of resistance depends on the size and type of the equipment, and voltage rating.

The insulation resistance is directly measured with the help of Insulation Resistance Tester which applies constant test voltage (d.c) on the insulation. This test voltage has to be applied for the time required for the absorption current to die down, (normally 1 minute). Electrical installations shall be tested for insulation resistance between all conductors and earth and between the conductors themselves.

As per the CEA Regulations 33:

- (i) all equipment shall have the insulation resistance (IR) value as stipulated in the relevant Indian Standards;
- (ii) on application of 500 V DC between each live conductor and earth for a period of one minute the insulation resistance of installation and equipment of voltage not exceeding 650 V shall be at least 1 MEGA OHM or as specified in the relevant Indian Standard;
- (iii) on application of 2.5 kV-DC between each live conductor and earth for a period of one minute, the insulation resistance, of installation and equipment of voltage exceeding 650 V but not exceeding 33 kV shall be at least 5 MEGA OHM or as specified in the relevant Indian Standard.

If the insulation resistance values obtained are less than required value, the approved method of drying out/ repairs shall be applied until the required steady insulation resistance value is obtained.

	OISD – STD – 137	Page No. 24
	INSPECTION OF ELECTRICAL EQUIPMENT	

II) MEASUREMENT OF EARTH- RESISTANCE

- a) Earth resistance can be directly read through an earth resistance tester which has associated Test, auxiliary Current and Potential electrodes. This instrument which is a combination of ohmmeter and generator works on 'fall of potential' principle. Test voltage is derived from the generator of the earth resistance tester.

Two auxiliary earth electrodes, besides the test electrode, are placed at suitable distance from the test electrode as shown in figure below. A measured current is passed between the electrode 'A' to be tested and an auxiliary current electrode 'C', and the potential difference between the electrode 'A' and auxiliary potential electrode 'B' is measured. The resistance of the test electrode 'A' is then given by:

$$R = \frac{V}{I}$$

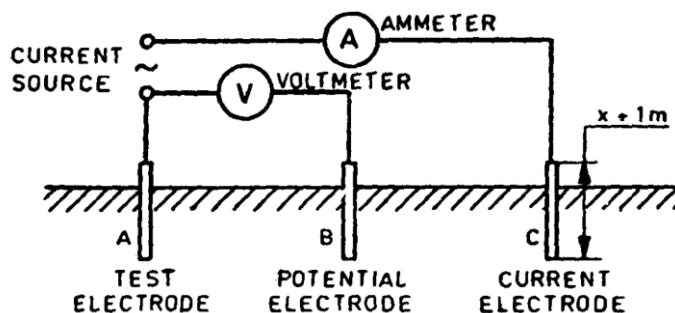
Where,

R = Resistance of the test electrode in ohms

V = Reading of the voltmeter in volts

I = Reading of the ammeter in amps.

All the three electrodes shall be so placed that they are independent of the resistance area of each other. If the test electrode is in the form of rod, pipe or plate, the auxiliary current electrode 'C' shall be at least 30m away from it, and the auxiliary potential electrode 'B' shall be placed mid-way between them.



Though the Fall of Potential is extremely reliable, highly accurate it has the drawback of being extremely time consuming, individual ground electrodes must be disconnected from the system to be measured and there are situations where disconnection is not possible. A clamp on earth tester performs the test in which the electrodes are not disconnected. The tester induces a known voltage in a loop circuit that includes ground, measures the resultant current flow and calculates the loop resistance. The limitations of the clamp-on tester are that it is only effective in situation with multiple grounds in parallel, it cannot be used on isolated grounds, as there is no return path.

Both the methods are acceptable in their appropriate application.

b) Allowable earth-Resistance Values

The resistance value of an earthing system to general mass of the earth should not exceed -

- For the electrical system and equipment, a value that ensures the operation of the protective device in the electrical circuit but not in excess of 4 ohms. However, for generating stations and large sub-stations this value shall not be more than 1 ohm.

	OISD – STD – 137	Page No. 25
	INSPECTION OF ELECTRICAL EQUIPMENT	

- For lightning protection, the value of 4 ohms as earth resistance shall be desirable, but in no case it shall be more than 10 ohms.
- 7 Ohms for storage tanks
- 1 Ohm for main earth grid, and bonding connections between joints in pipelines & associated facilities.
- 2 Ohms for each electrode to the general mass of the earth^{Note}

Note:

1. In areas with high soil resistivity (e.g. quartzite, granite, lateritic soil etc.), individual pit resistance value up to 5 ohm is acceptable, but grid resistance shall not be more than 1 ohm.
2. In exceptional cases, where the individual pit resistance of less than 5 ohm is not achievable, number of rods/pipes/plates may be connected in parallel to achieve a combined pit resistance of 5 ohms. In this case also grid resistance shall not be more than 1 ohm.
3. Typical diagrams of resistances to general mass of earth are provided at Annexure-V
4. The electrical earthing lay out diagram with earth pit numbers need to be prepared, approved and should be kept available.

III) EARTH CONTINUITY TEST

All protective and bonding conductors must be tested to ensure that they are electrically safe and correctly connected. Method for measuring resistance of protective conductor by using the neutral conductor as return lead is as follows:

A temporary link is made at the distribution board between neutral and protective conductor systems (Don't forget to remove the link after testing). The low resistance tester is then connected to the earth and neutral of the point from which the measurement is taken (see {Fig below}). This gives the combined resistance of the protective and neutral conductors back to the distribution board.

Then

$$R_p = R \times \frac{A_n}{A_n + A_p}$$

where

R_p - is the resistance of the protective conductor

R - is the resistance reading taken

A_n - is the cross-sectional area of the neutral conductor

A_p - is the cross-sectional area of the protective conductor.

This method is only valid if both conductors have the same length and both are copper; in most cases where steel conduit or trunking is not used as the protective conductor, the test will give correct results.

	OISD – STD – 137	Page No. 26
	INSPECTION OF ELECTRICAL EQUIPMENT	

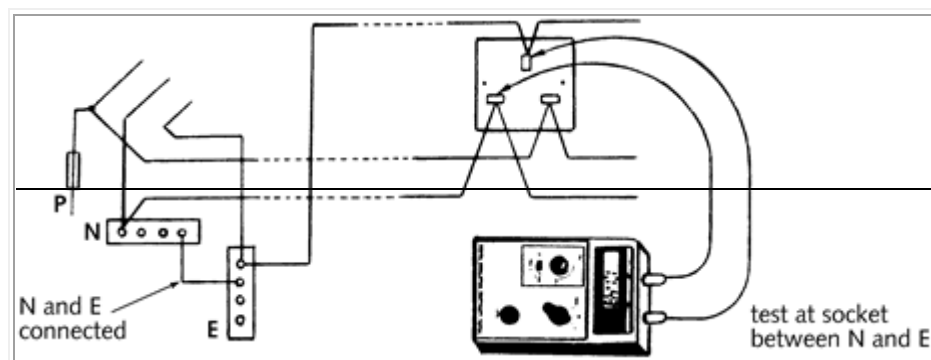


Fig : Protective conductor continuity test using the neutral conductor as the return lead

IV) POLARISATION INDEX TEST

The polarization index is a specialized application of the dielectric-absorption test. This test is used to determine the insulation resistance level of cable, transformers, motors and similar electrical equipment. Using a power driven insulation tester, the test voltage is applied to the equipment for a period of 10 minutes. The insulation resistance values at the end of 1 minute and 10 minutes period are recorded. The ratio of the final resistance to initial resistance is called the Polarisation index.

A fair index is greater than 2.0; a poor to questionable index is 2.0 to 1. When index level starts at 2.5 to 4 base level and begins to fall one or two years later to 1.5 or 2 it is apparent that either the insulation/ cable has absorbed moisture, or the insulation has become weak to an unsafe level.

The recommended values of PI for Rotating Machines based on IEEE Std. 43-2000 are as follows:

Thermal Class Rating (Insulation Class as per IEC 60085-01 :1984)	Minimum PI
Class A	1.5
Class B	2.0
Class F	2.0
Class H	2.0

Note: If the 1 min insulation resistance value is above 5000 MΩ, the calculated PI may not be meaningful. In such cases, the PI may be disregarded as a measure of winding condition. These values shall be considered only relative and are used to know the trend of insulation deterioration over a period of time.

V) MOISTURE ABSORPTION TEST (DIELECTRIC ABSORPTION RATIO TEST)

Moisture absorption is usually evaluated from the ratio of two values of insulation resistance measured with Insulation tester after 60 seconds and 30 seconds respectively. This ratio is called the Coefficient of moisture (dielectric) absorption, denoted by 'K'.

$$K = \frac{R_{60}}{R_{30}}$$

Damp installation will have a coefficient value close to 1. As the insulation dries out, the coefficient rises and when it becomes 1.3 or greater the insulation is considered to be dry.

	OISD – STD – 137	Page No. 27
	INSPECTION OF ELECTRICAL EQUIPMENT	

The following table can be used as reference (Ref Table 20.20.2 of NFPA 70B):

Condition	60:30 Second Ratio	10:1 Min Ratio
Dangerous	-----	Less than 1
Poor	Less than 1.1	Less than 1.5
Questionable	1.1 to 1.25	1.5 to 2
Fair	1.25 to 1.4	2 to 3
Good	1.4 to 1.6	3 to 4
Excellent	Above 1.6	Above 4

VI) CATHODIC PROTECTION (ELECTRICAL TEST)

The underground or submerged structures such as pipelines, mounded bullets shall be protected against electrolytic corrosion by cathodic protection, either by the “impressed current” or the “sacrificial anode” methods.

Periodic testing of cathodic protection system such as structure to soil voltage, anode current, soil resistivity, performance of transformer and rectifier shall be carried out to ensure continued protection.

Current output and voltages shall be adjusted to maintain the degree of cathodic protection.

Allowable values of structure to soil voltage shall not be less than (-) 0.85 volts with respect to copper /copper sulphate half cell. Current density requirement depends on the condition of coating and soil resistivity. Better the quality of coating, lower will be the requirement of current. The upper limit for the PSP shall not exceed (-) 1.2 V to avoid disbonding of insulation wrap coating.

VII) CABLE FAULT DETECTION

Annexure IV describes one of the procedures for detection of cable fault location, which shall be used as a guideline if specific instruction from the cable manufacturer is not available.

VIII) TESTS ON LIGHTNING PROTECTION SYSTEM

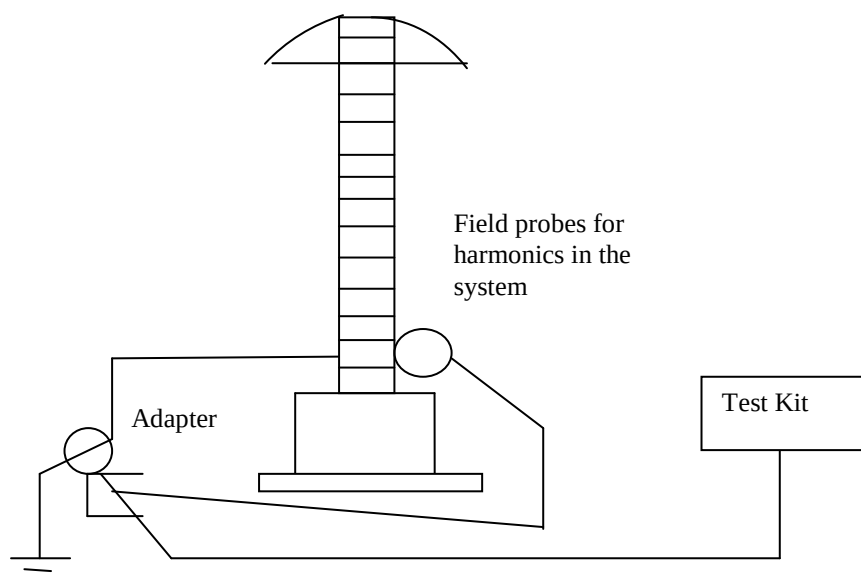
- Building and Structures: The ohmic resistance of lightning protective system complete with air terminations but excluding earth electrode connections should not exceed 1 ohm. The continuity of lightning conductors and bonding connections of structures shall be tested to ensure that there are no open circuits. Also, checks shall be made to ensure that the clearances from other structures are maintained to prevent flashover to the structure during lightning.
- Lightning arrester: On Line Measurement of Third Harmonic Resistive Leakage Current.

The total current (IT) through the Surge Arrester having Capacitive component (IC) and Resistive Current (IR) flows through the Leakage Current monitor. The IR component has 3rd, 5th and other harmonics present in it. When degradation of the metal oxide discs occur, the variation of 3rd harmonic Resistive currents is more pronounced than other harmonics currents, total current (IT) and total Capacitive current (IC).

Test Procedure for Online measurement of Third Harmonic Resistive Current:

	OISD – STD – 137	Page No. 28
	INSPECTION OF ELECTRICAL EQUIPMENT	

Typical connections are shown in figure below -



A clamp type CT probe is connected to the ground wire which allows the leakage current to be detected. The probe output passes through a shielded cable to the detector unit. A band pass filter is built into the initial stage to pass the 3rd harmonic and to cut off the 5th & higher harmonics. After the 3rd harmonic is selected, it is amplified and then displayed. A compensating antenna is also provided, output of which is fed to the test kit to compensate for the harmonics present in the system voltage. The harmonics present in the EHV system are compensated and filter circuits are provided for bypassing undesired currents and only 3rd harmonic resistive current is measured.

The following parameters are measured:

- True RMS value of total current, flowing through ground circuit.
- Peak value of total current
- True RMS value of third harmonic leakage current.
- Ambient Temperature.

Interpretation of Test results:

The non-linear voltage-current characteristic of a metal-oxide arrester gives rise to harmonics in the leakage current when the arrester is energized with a sinusoidal voltage. The harmonic content depends on the magnitude of the resistive current and the degree of non-linearity, which is a function of voltage and temperature. The third harmonic content of the resistive current is typically 10 % to 40 % of the Total Resistive current.

An increase in the third harmonic resistive component of leakage current brings the arrester to thermal overloading and finally causes breakdown. If the increase in the 3rd harmonic current is monitored and detected earlier, then arrester can be repaired or replaced, thus preventing damage.

The parameters of currents recorded are compared with previous measured values on the same arrester. The values are also compared with arrester of similar make and type, operating under similar conditions.

	OISD – STD – 137	Page No. 29
	INSPECTION OF ELECTRICAL EQUIPMENT	

Acceptable limits of Third Harmonic Resistive Current as per CBIP Manual on EHV Substation Equipment Maintenance (Publication No. 294) (Refer Annexure -1, Table 11.9)

Table 1:

Life of LAs	Acceptable value of THRC	Remarks
New LA	Upto 20 μ A	Normal
In service LA	Upto 150 μ A	Normal
	150 to 350 μ A	To be tested for IR & if found low, to be removed from service
	>350 μ A for Gapless type	To be removed from service
	>500 μ A for Gapped type	To be removed from service

IX) TESTING OF TRANSFORMER

i) **Tan Delta Testing (Power Factor and dissipation Factor)** - for Power and Distribution Transformers

The dielectric loss in an insulation system is the power dissipated by the insulation when subjected to an applied alternating voltage. Good insulation usually has a very low loss, while a high loss may indicate problems in the insulation structure.

A dielectric loss testing program provides several important benefits. Periodic tests performed during the service life of the equipment can indicate that the insulation is either aging normally or deteriorating rapidly.

The dielectric loss is usually determined by a bridge measuring instrument, such as the Schering Bridge. The test can be performed at any voltage within the normal operating range of the equipment under test, however usually performed at rated voltage or a maximum of 10kV.

A visual inspection of the apparatus should be performed to identify external damage or unusual conditions. The instruction of the test kit should be strictly followed.

The recommended value of Power Factor as per IEEE Std. 62-1995 is <0.5% for New Transformer and should be < 2% for service aged transformers.

Additional specialized tests like the Partial Discharge and Sweep frequency response analysis (SFRA) may also be conducted to assess the health of the transformer.

ii) **Partial Discharge Testing** - for Instrument Transformers (Wound Type)

Partial Discharge testing provides a good assessment of the insulation condition of the Instrument Transformers. Partial Discharge testing should be done for equipment having solid insulation and the highest equipment voltage is 7.2 kV and above and for equipment having liquid immersed insulation and equipment voltage 72.5 kV and above.

The testing shall be done as per IS 11322 (Method for Partial Discharge Measurement in Instrument Transformers) and shall be carried out after all dielectric tests are completed. The applied voltage shall be raised to the pre-stress voltage before reducing it to the partial discharge measuring voltage. The permissible partial discharge levels are given in the table below:

	OISD – STD – 137	Page No. 30
	INSPECTION OF ELECTRICAL EQUIPMENT	

Type of Earthing	Type of Instrument Transformer	Pre-Stress Voltages	Measuring Voltage	Type of Insulation	Permissible Partial Discharge Level
Networks with an isolated or resonant earthed star point	Current Transformer	$1.3 U_m$	$1.1U_m$	Liquid Immersed	100 pC
				Solid	250 pC
	Phase to Earth Voltage Transformer	$1.3 U_m$	$1.1U_m/\sqrt{3}$	Liquid Immersed	10 pC
				Solid	50 pC
	Phase to Phase Voltage Transformer	$1.3 U_m$	$1.1U_m$	Liquid Immersed	10 pC
				Solid	50 pC
Network with an effectively earthed starpoint	Current Transformer	$0.8 \times 1.3 U_m$	$1.1U_m/\sqrt{3}$	Liquid Immersed	10 pC
				Solid	50 pC
	Phase to Earth Voltage Transformer	$0.8 \times 1.3 U_m$	$1.1U_m/\sqrt{3}$	Liquid Immersed	10 pC
				Solid	50 pC
	Phase to Phase Voltage Transformer	$1.3 U_m$	$1.1U_m$	Liquid Immersed	10 pC
				Solid	50 pC

iii) Transformer Oil Testing

Proper care shall be taken for taking oil samples for testing. It should be ensured that the sample bottle is clean and dry. The bottle shall be properly rinsed with the same oil which has to be tested, before filling up the sample. The Oil Testing shall be carried out as per IS 1866. The various tests and the recommended limits for the characteristics of Oil based on IS 1866 are as follows:

Property	Highest Voltage of Equipment (kV)		
	<72.5	72.5 to 170	>170
Appearance	Clear, free from sediment and suspended matter		
Density at 29.5°C (g/cm ³), Max	0.89	0.89	0.89
Viscosity at 27°C (cSt), Max	27	27	27
Flash Point, (°C), Min	140	140	140
Pour Point, (°C), Max	-6	-6	-6
Neutralization Value (mg KOH/g), Max	0.03	0.03	0.03
Water Content (ppm), Max	20	15	10
Interfacial Tension (mN/m), Min	35	35	35
Dielectric Dissipation Factor at 90°C and 40 Hz to 60 Hz.	0.015	0.015	0.010
Resistivity (90°C) $\times 10^{12}$ (Ohm-cm), Min	6	6	6
Breakdown Voltage (kV), Min	40	50	60
Oxidation stability of uninhibited oil			
i) Neutralization Value	0.4	0.4	0.4

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	OISD – STD – 137	Page No. 31
	INSPECTION OF ELECTRICAL EQUIPMENT	

(mg KOH/g), Max ii) Sludge (percent by mass), Max	0.1	0.1	0.1
Oxidation stability for inhibited oil Induction period (hours)	Similar values as before filling		

As per IS 1866/2000 testing frequency for Transformer oil testing for various parameters are as below.

Property	Frequency of Testing
Appearance	In conjunction with other Quantitative tests
Water content	After filling or refilling prior to energizing, then after three and 12 months, subsequently along with DGA
BDV	After filling or refilling prior to energizing, then yearly
Total acidity	Yearly
IFT	After filling or refilling prior to energizing, then yearly
Resistivity	After filling or refilling prior to energizing, then yearly
Tan δ	After filling or refilling prior to energizing, then yearly
Flash Point	Yearly
Sediment and sludge	Yearly

iv) DISSOLVED GAS ANALYSIS (DGA)

This test is used to determine the amount of specific gases generated in Transformer oil during service. The amount of gases and certain combinations of those is the first indication of a possible malfunction that may eventually lead to failure if not corrected. Some of the mechanisms which lead to generation of gases due to the chemical decomposition of oil are arcing, partial discharges, low-energy sparking, severe overloading, overheating of the insulation etc. Certain empirical values are provided to give an indication of the impending failure.

For DGA testing the sample should be obtained using a clean, moisture free, gas tight container. Care should be taken to purge the container of all free gas at the time of taking the sample.

The first step in evaluating DGA results is to consider the concentration levels (in ppm) of each key gas. It is recommended that values for each of the key gases be trended over time so that the rate-of-change of the various gas concentrations can be evaluated. Basically, any sharp increase in key gas concentration is indicative of a potential problem within the transformer. Below is a table which has been derived from ANSI/IEEE C57.104 information. The suggested action levels for key gas concentrations.

	OISD – STD – 137		Page No. 32
	INSPECTION OF ELECTRICAL EQUIPMENT		

Gas Description		Normal Limits* (<)	Action Limits** (>)	Potential fault Type
Hydrogen	H ₂	150	1000	Corona, Arcing
Methane	CH ₄	25	80	Sparking
Acetylene	C ₂ H ₂	15	70	Arcing
Ethylene	C ₂ H ₄	20	150	Severe overheating
Ethane	C ₂ H ₆	10	35	Local Overheating
Carbon Mono Oxide	CO	500	1000	Severe overheating
Carbon dioxide	CO ₂	10000	15000	Severe overheating
Total Combustibles.	TDCG	720	4630	
* As the values exceed this limit, sample frequency should be increased with consideration given to planned outage in near term for future evaluation.				
** As value exceeds this limit, removal of transformer from service should be considered.				

The test should be performed in a laboratory and ASTM D 3612-93 should be followed for extraction and analysis of the key dissolved gases. Refer IS 9434 for the sampling and testing procedure.

Interpretation:

Based on Key Gas Method (IEEE PC 57.104 D11d)

KEY GAS	FAULT TYPE	TYPICAL PROPORTIONS OF GENERATED COMBUSTIBLE GASES
C ₂ H ₄	Thermal Oil	Mainly C ₂ H ₄ . Smaller proportions of C ₂ H ₆ , CH ₄ and H ₂ . Traces of C ₂ H ₂ at very high fault temperatures.
CO	Thermal Oil and Cellulose	Mainly CO. Much smaller quantities of hydrocarbon gases in same proportions as thermal faults in oil alone.
H ₂	Electrical Low Energy Partial Discharge	Mainly H ₂ . Small quantities of CH ₄ , Traces of C ₂ H ₄ and C ₂ H ₆ .
H ₂ & C ₂ H ₂	Electrical High Energy (Arcing)	Mainly H ₂ & C ₂ H ₂ . Minor traces of CH ₄ , C ₂ H ₄ , and C ₂ H ₆ . Also CO if cellulose is involved.

Rogers Ratio (IEEE PC 57.104 D11d)

Ratio 1	Ratio 2	Ratio 3	Suggested Fault Type
CH ₄ /H ₂	C ₂ H ₂ /C ₂ H ₄	C ₂ H ₄ /C ₂ H ₆	
<0.1	<0.01	<1.0	Case 0: Normal
≥ 0.1, <0.5	≥1.0	≥1.0	Case 1: Discharge of low energy
≥ 0.1, < 1.0	≥0.6, <3.0	≥2.0	Case 2: Discharge of high energy
≥1	<0.01	<1.0	Case 3: thermal fault, low temperature < 300°C
≥1	<0.1	≥1.0, < 4.0	Case 4: Thermal Fault, < 700 °C
≥1	<0.2	≥4.0	Case 5: Thermal Fault, > 700 °C

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	OISD – STD – 137	Page No. 33
	INSPECTION OF ELECTRICAL EQUIPMENT	

Additional test like the Furan analysis may also be conducted to assess the condition of the insulation paper, and based on the furan analysis a degree of polymerization test can be carried out on the insulation paper sample.

X) VLF TESTING OF CABLES

There are two types of tests for cables viz. Maintenance Tests and Diagnostic Tests or Condition Assessment Tests.

- i) **Maintenance Test** – Is a field test made during the operating life of a cable system. It is intended to detect deterioration of the cable system and to check the serviceability so that suitable maintenance procedures can be initiated.

Recommended Test Voltages for withstand test (Based on IEEE 400.2-2004):

Suggested field test voltages with 0.1 Hz VLF Voltage Source for shielded power cables **(based on IEEE 400.2- 2004)** is given in table below

VLF test voltages for sinusoidal waveform

Cable rating rms voltage in KV (ph-ph)	Voltage for Acceptance tests rms voltage in kV (ph – earth)	Voltage for Maintenance tests rms voltage in kV (ph – earth)
6.6	8	6
11	13	10
22	25	20
33	38	30

Suggested duration for Maintenance test voltage is **15 min.**

ii) **Condition assessment / Diagnostic Tests:**

There are two diagnostic tests viz. Dissipation Factor and Partial Discharge Test.

1. Dissipation factor (Tan δ) tests.

The Dissipation factor (**Tan δ**) is a diagnostic test for the evaluation of the cable insulation at operating or Test voltage levels. The quantity Tan δ / Dissipation factor is a measure of the losses of the insulating dielectric in an AC electric field.

Recommended Test Voltages:

The Dissipation factor is measured at 0.1 Hz with a test Voltage of V_0 , where V_0 is the nominal operating voltage (Ph-E). The voltages may be applied in steps up to V_0 and the trend of Tan δ values may be plotted. The Tan δ values at higher voltages are compared to those at lower voltages and an analysis is done. If a cable's insulation is perfect, the Tan δ value will not change as the applied voltage increases. Values over a period of time are compared to assess the condition of the cable.

	OISD – STD – 137	Page No. 34
	INSPECTION OF ELECTRICAL EQUIPMENT	

Interpretation of Test results:

The acceptable values of $\tan \delta$ for XLPE/ PILC cables at V_0 (as per IS 7098 -Part 2) are as mentioned below.

Tan δ at V_0 (at ambient temperature)	$\Delta \tan \delta$ between 0.5 V_0 and 2V_0	Interpretation
$< 4.0 \times 10^{-3}$	$< 2.0 \times 10^{-3}$	Condition of cable insulation is good
$> 4.0 \times 10^{-3}$	-	Cable needs to be replaced within 6 months

In this test, voltage is to be increased in steps. While increasing the voltage if $\Delta \tan \delta$ between 0.5 V_0 and the applied voltage is $\geq 2.0 \times 10^{-3}$, cable is to be considered as aged and testing to be stopped.

Through the Dissipation factor test, the Cable insulation system can be monitored over time and a cable system history can be developed.

2. Partial discharge measurement for cables

Note: The PD testing shall be carried out only after VLF Tan delta values are acceptable.

Partial Discharge measurement is a sensitive method of detecting small imperfections in the insulation such as voids or skips in the insulation shield layer. Partial Discharge measurement tests can be done either as On-line or Off-lines tests. It is recommended to adopt the Off-line PD test method.

The Voltage Sources for Off-line PD measurements can either be **Power frequency** voltage source or a **Very Low frequency Voltage** source. However to be able to test at higher voltage VLF kit is utilized as its VA rating will be less than power frequency test kit.

Recommended Test Voltages:

The applied test voltages for PD measurements in XLPE/ PILC-cables should be as follows, as recommended by **IEEE -400.3-2006**

Cable type	Applied Voltage
New Cables	Max 3 V_0 (with the concurrence of the cable manufacturer)
Service aged cables	Max 2 V_0

V_0 – nominal operating voltage (Ph-E)

The applied voltage is to be raised in steps starting from 0.5 V_0 .

	OISD – STD – 137	Page No. 35
	INSPECTION OF ELECTRICAL EQUIPMENT	

Interpretation of Test results:

Although there are no prescribed values for Tolerance limits recommended by Standards, based on the IEEE 400.3 – 2006 testing, the results may be interpreted as below based on the PD activity, PD Inception Voltage (PDIV) and PD Extinction Voltage (PDEV)

Result	High PD *	Low PD *	Remarks
PDEV < 1Vo	Bad cable	Bad cable	Plan for replacement immediately
PDIV < 1Vo	Bad cable	Bad cable	Plan for replacement immediately
PDIV < 1.3 Vo & PDEV > 1 Vo	Bad cable, plan for replacement immediately	Good cable Re-test after 12 months	
PDIV > 1.3 Vo & PDEV > 1 Vo	Some Concern. Re-test after 6 months	Good cable Monitor Re-test after 24 Months	
PDIV > 2 Vo & PDEV > 1 Vo	Some Concern. Re-test after 24 Months	Very Good Cable	

* Reference High and Low PD Values are provided in the table below.

However, in order to achieve accurate interpretation of test results, it is required to compare with earlier test data and monitor trends.

Reference Table for PD Values:

There are no standards for acceptable values for PD. Based on certain white papers the values given in the following tables can be used as reference. These values provide a basis for developing a database of in-service cables based on their insulation condition. It is recommended that each site develop their own reference tables to assess the risk.

Type of Cable	PD Value in pC	Remarks	PD Category
PILC	0-3000	Discharge within acceptable limits.	Low
	3000- 6500	Some Concern, monitoring recommended	
	6500-10000	Some Concern, regular monitoring recommended	High
	> 10000	Major Concern, repair or replace.	
XLPE	0-250	Discharge within acceptable limits.	Low
	250-350	Some Concern, monitoring recommended	

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	OISD – STD – 137	Page No. 36
	INSPECTION OF ELECTRICAL EQUIPMENT	

Type of Cable	PD Value in pC	Remarks	PD Category
	350-500	Some Concern, regular monitoring recommended	High
	>500	Major Concern, repair or replace.	

As discussed above, each of the diagnostic tests have their merits and de-merits. The suitability, practicality and effectiveness of these tests at Site depend on various factors such as the age and type of the insulation, the voltage level required to detect a defect and availability of test equipment.

Meaningful interpretation of the test results will require development of a database to better define acceptance criteria.

XI) THERMOGRAPHY:

Heat is often an early symptom of equipment damage or malfunction for electrical equipment. Checking the temperature of electrical equipment on a regular basis helps in quickly identifying unusual temperature conditions and reduces the likelihood of unplanned downtime due to equipment failure.

In thermography the thermal image of the equipment is taken, the infrared image captures the equipment temperature profile.

Thermography can be used to measure bearing temperatures in large motors, hot spots in electrical switchgear, identify faulty terminations, and identify electrical equipment like transformers, LA, circuit breakers operating at high temperatures.

It is recommended that each organization should develop a thermography program for electrical equipment.

For additional details ASTM E1934 (Standard Guide for examining electrical and mechanical equipment with infrared thermography) may be referred.

As per the Maintenance Testing Specifications for Electric Power Distribution Equipment and Systems – International Electric Testing Association (NETA) the Delta T Criteria for Electrical System is as follows:

Priority	Delta T between similar components under similar load	Delta T over ambient temperature	Recommended Action
4	1 to 3 Deg. C	1 to 10 Deg. C	Possible deficiency; warrants investigation.
3	4 to 15 Deg. C	11 to 20 Deg. C	Indicates probable deficiency; repair as time permits.
2	-----	21 to 40 Deg. C	Monitor until corrective measures can be accomplished
1	> 15 Deg. C	> 40 Deg. C	Major discrepancy; repair immediately

	OISD – STD – 137	Page No. 37
	INSPECTION OF ELECTRICAL EQUIPMENT	

8.0 DOCUMENTATION

Documentation and record keeping form an integral part of total inspection management system. The various records, which shall be maintained in any unit, are summarised below:

8.1 FAILURE ANALYSIS REPORTS

Failure analysis shall be conducted on

- i) High Voltage equipment
- ii) Repetitive failures of any kind
- iii) Power system failure affecting plant operation
- iv) Failure due to arcing or sparking, which caused unsafe condition/accidents.
- v) Failure to newly installed or repaired systems
- vi) Critical equipment and systems.

For this, various data and records shall be regularly maintained through Failure analysis reports (in the formats suggested for few, as per Annexure II), and monitored by Electrical in-charge.

8.2 HISTORY CARDS

History cards for following equipment shall be maintained by each Unit-in-charge or Maintenance Planning Cell, which should be the basis for carrying out failure analysis and enabling replacement with identical equipment or components during emergencies:

- (i) Rotating equipment (Motors, Generators etc.)
- (ii) Stationary equipment (Switchboard, Transformers, Reactors, Battery charger, UPS, switchyard etc.)
- (iii) Protective system (Protective relays, Earthing, Cathodic protection etc.)
- (iii) Distribution network (Cables, Overhear lines etc.)
- (iv) Other critical electrical equipment and systems

Effort should be made to maintain the equipment design data, commissioning data, maintenance & failure data in soft form. This will help in easy retrieval and analysis.

Formats for few of the above have been suggested at Annexure - I

8.3 TEST RECORDS AND OTHER DATA

Records for the following tests and other data listed shall be available:

- i) Earth resistance tests
 - a. Tanks, gantry structures & process area
 - b. Vessels/exchangers/columns
 - c. Motors/generators
 - d. Earthing pits
 - e. Bonding and grounding of loading arms and piping
 - f. Substations/switchyards
 - g. Lightning protection system

	OISD – STD – 137	Page No. 38
	INSPECTION OF ELECTRICAL EQUIPMENT	

- ii) Insulation resistance tests of cables, motors, transformers and other critical electrical equipment.
- iii) Cathodic protection parameters (potentials at identified locations of tank bottoms, structures/ pipelines etc.)
- iv) Transformer oil tests
- v) Calibration tests of meters / instruments
- vi) Simulation test on protective relay circuits.
- vii) Relay test records
- viii) Test records on battery
- ix) Other Records
 - a) Compliance reports of previous audit reports and Electrical Inspector's observations during periodical statutory inspections under CEA Regulation 30 (for the last 5 years)
 - b) Factory acceptance test certificates, Commissioning & field test records of critical equipment before energisation and after subsequent repairs.
 - c) Updated equipment list (load summary)
 - d) Updated single line diagrams and control schematics.
 - e) Communication and fire alarm system
 - f) Line clearance records.
 - g) List of designated persons. (Annexure III)
 - h) Reports on investigation of accidents (for the last 5 years), if any
 - i) Specifications for spare parts (for ordering purposes) with original Manufacturer's catalogues and descriptive technical data.
 - j) Operation and maintenance instruction manuals
 - k) Site specific SOPs and SMPs.
 - l) Reference documents (International and Indian Standards, CEA Regulation 2010, OISD standards and latest statutory regulations)
 - m) Relay co-ordination drawings and table of relay settings.
 - n) Hard copies of the relay settings & logics downloaded from Numerical relay should be signed and documented.
 - o) Hard copies of protection logic and control set points downloaded from DCS/ PLCs should be signed and documented.
 - p) Substation log register with updated entries and observations.

	OISD – STD – 137	Page No. 39
	INSPECTION OF ELECTRICAL EQUIPMENT	

9.0 PRECOMMISSIONING CHECKS

Before commissioning electrical equipment after repairs and maintenance, the following minimum Pre-commissioning checks shall be carried out. Wherever manufacturer's recommendations are available, the same shall prevail.

In the case of newly installed facilities also, the same Pre-commissioning checks shall be applicable but additionally the following shall be ensured:

- I) The statutory clearances from the State/ Central Electrical Inspectorate, Directorate General of Mines Safety, Petroleum and Explosives Safety Organisation, others statutory bodies (as applicable) shall be obtained.
- II) Manufacturer's test certificates shall be obtained (for the successful passing of shop tests.
- III) Installations conform to acceptable engineering standards and are appropriate to the respective hazardous area classifications. Installations with all individual Isolating devices (for safe disconnection of electrical circuits) shall be easily accessible for regular maintenance and emergency operation.

9.1 ROTATING MACHINERY

9.1.1 Generators

The following items shall be checked/tested, before start up:

- i) Check Name plate details.
- ii) External body of generator for any damage.
- iii) Checking of Exciter/ Field/ Stator coils, Brushes (condition, spring pressure).
- iv) Testing of AVR/ DVR including rotating rectified assembly and its components.
- v) Checking of Safety Interlocks.
- vi) Primary injection testing of generator panel.
- vii) Secondary injection testing of protection relays.
- viii) Check protection relays are set as per recommended settings.
- ix) Checking of Annunciation circuit.
- x) Condition of terminal box: missing/ loose nuts & bolts.
- xi) Foundation bolts: tightness
- xii) Cable connections
- xiii) Blockage of ventilator openings
- xiv) Condition of bearings and insulation resistance
- xv) Oil lubrication (adequacy & Oil quality)
- xvi) Alignment of coupled machines
- xvii) Healthiness of space heaters.
- xviii) Measure & record Insulation resistance & Polarisation Index values.
- xix) Earthing connections (body & neutral)
- xx) Protection circuits (simulation tests)
- xxi) Auto/manual starting and changeover system
- xxii) Checking of synchronising panel
- xxiii) Generator set(s) have nomenclature such as G-1, G-2, Set-1 or Set-1 for ease of identification.
- xxiv) **In case of Acoustic Enclosure**
Check the following:
 - a. Measure the noise level to ensure that it is as per CPCB norms. (Max 70db within 1 metre the D.G package in any direction as per CPCB norms)
 - b. Adequacy of Illumination inside the enclosure
 - c. Functionality of Gas Detector.

	OISD – STD – 137	Page No. 40
	INSPECTION OF ELECTRICAL EQUIPMENT	

- d. Functionality of Emergency Shut Down Switch which shall be outside the enclosure.
- e. Functionality of Blower fans.
- xxv) Checks/ testing as recommended by OEM.

The following items shall be checked after start up:

- i) Voltages building up in all the phases and also phase sequence.
- ii) Exciter voltage
- iii) Exciter current
- iv) Load on the generator
- v) Testing of speed and voltage raise/ lower operation from control panel
- vi) Overheating of alternator/exciter/bearings
- vii) Speed whether steady or hunting and operation of AVR.
- viii) Sparking of brushes
- ix) Commutator surface for uniformity (projection, if any)
- x) Carbon dust collection (whether consolidated between the commutator segments)
- xi) Audio visual annunciation
- xii) Noise level/ any abnormal sound
- xiii) Brush spring pressure.

9.1.2 Motors and Controlgear

The items listed below shall be checked/ tested before start up:

- i) Check name plate details.
- ii) Ensure motor 'type' is as per approved area classification drawings.
- iii) Functional checks of motor control and protective circuits with interlocks and all audio visual annunciation (through simulation test)
- iv) Checking of Annunciation circuit.
- v) Check protection relays are set as per recommended settings.
- vi) Condition of terminal box: missing/ loose nuts & bolts.
- vii) Foundation bolts: tightness
- viii) Measure & record Insulation resistance & Polarisation Index values.
- ix) Cleanliness (free from dirt, dust and moisture)
- x) Insulation resistance (before and after connection of the power cables)
- xi) Polarisation Index and Dielectric Absorption tests (for critical equipment only)
- xii) Sliprings, brushgear polish and cleanliness of machine winding.
- xiii) Brushes in holders (check for clearance and pressure)
- xiv) Connection and operation of space heaters, where provided. (ensure its automatic switching off when motor is started and switching on when motor is stopped)
- xv) Provision of emergency stop switches and coupling guard
- xvi) External damages, and missing components
- xvii) Freeness of rotation of motor shaft, and lubrication.
- xviii) Cable termination (check also for proper size and type of cable)
- xix) Ratings of power & control fuses are as per approved drawings. And ensure -thermal, short circuit & earth fault settings (protective system)
- xx) Earthing connections/earth resistance
- xxi) Flameproofness, corrosion and weather protection as per area classification.
- xxii) Voltage drop
- xxiii) Alignment with coupled machine.
- xxiv) Checks/ testing as recommended by OEM.

The items listed below shall be checked after start up:

- i) Healthiness of Cooling system
- ii) No load run of motors with observations for
 - a. Supply voltage and frequency
 - b. Direction of rotation.

	OISD – STD – 137	Page No. 41
	INSPECTION OF ELECTRICAL EQUIPMENT	

- c. Abnormal sound
- d. Temperature
- e. Vibration
- f. No load current
- g. Speed

Note:

Run the uncoupled motor for a minimum period of 4 hours for LT Motors and 8 hours for HT Motors and ensure trouble free operation.

9.2 STATIONARY EQUIPMENT /SYSTEMS

9.2.1 Switchboards & Panels

The following shall be checked/tested:

- i) Tightness of foundation bolts.
- ii) Check enclosures and cable boxes of incomer, bus-coupler and all outgoing feeder for any foreign object, condition of support insulators, phase barriers.
- iii) Switchgear earthing by two separate and distinct earthing connections.
- iv) Primary injection testing of switchgear Incomer and Outgoing panels.
- v) Secondary injection testing of protection relays.
- vi) Check protection relays are set as per recommended settings.
- vii) Functionality check/ testing of CBs, MCCBs, Energy meter, Power packs etc.
- viii) Operation of interlocks and safety devices
- ix) Relay and meter calibrations (simulation)
- x) Check for proper cabling and termination.
- xi) Cleanliness (free from dirt, dust, moisture etc.)
- xii) Check for any tool or foreign material inside the panel
- xiii) All circuit wiring, terminations intact, and ferrule numbers checked.
- xiv) Tightness of connections/clamping of wires, and tags of all the components.
- xv) Vermin proofness (whether unused openings closed)
- xvi) Simulation of Protection and annunciation circuits including checking of relay settings, and fuse ratings.
- xvii) Provision of caution notices, tags, and danger board
- xviii) Provision of insulating mats conforming to IS:15652:2006 having ISI mark
- xix) Continuity checks of wires
- xx) Insulation resistance.
- xxi) Conformity with updated schematic and single line diagrams (ratings/settings)
- xxii) Timer checks
- xxiii) Short in CTs, blockage in relays
- xxiv) External damage and dents
- xxv) Cable bends within the switchboard (whether required bending radius is maintained)
- xxvi) Contact alignment of draw out type switchgear
- xxvii) Oil level, and oil condition for the oil-immersed gear
- xxviii) Healthiness of space heaters.
- xxix) Control supply availability
- xxx) Provision of fire fighting equipment suitable for electrical fires.
- xxxi) Electrical & manual closing and tripping of all circuit breakers (remote and local)
- xxxii) Provision of phase barriers and partitions (to avoid inadvertent contact with live parts)
- xxxiii) Condition of supporting structure.
- xxxiv) Earthing of panel door with the panel.
- xxxv) Checks/ testing as recommended by OEM.

	OISD – STD – 137	Page No. 42
	INSPECTION OF ELECTRICAL EQUIPMENT	

9.2.2 Instrument Transformers (CT's and PT's)

The following checks/tests shall be carried out:

- i) Polarity
- ii) Continuity
- iii) Ratio error
- iv) Tightness of wiring connections
- v) Wiring (continuity and completeness)
- vi) Checks/ testing as recommended by OEM.

9.2.3 Transformer & Busduct

The following checks/tests shall be carried out:

- i) Check name plate details
- ii) Provision of all the accessories
- iii) Primary injection testing of Transformer panel.
- iv) Secondary injection testing of protection relays.
- v) Testing of transformer winding/ oil alarm/ trip ckts, buchholz relay operation.
- vi) External damages (especially on bushings, cooling fins, radiators)
- vii) Insulation resistance & polarisation index testing of HV and LV windings.
- viii) Body and neutral earthing/earth resistance
- ix) Dielectric strength testing of Transformer oil (Conforming to IS:335)
- x) Check for any oil leakage.
- xi) Oil level in the conservator, bushings and thermometer pockets
- xii) Condition of silica gel.
- xiii) Blockage, if any, in the breather pipe
- xiv) Explosion vent diaphragm not having any dents, accumulation of any oil/air (whether released)
- xv) Buchholz relay (frictionless movement and floats)
- xvi) Protective checks on trips and alarms (simulation)
- xvii) Arcing gaps for the bushings
- xviii) Earthing connection for the surge diverter
- xix) Operation of the tap changer circuits on all the tap positions (simulation)
- xx) Identical indication of tap position on transformer and on control panel
- xxi) Operation of tap changer limit switch on the maximum tap positions.
- xxii) Oil condition and oil level in tap changer tank
- xxiii) Insulation resistance of all control circuits and transformer windings.
- xxiv) Cooling system (automatic starting and stopping of fans)
- xxv) Cleanliness of transformer surface
- xxvi) Space heater operation (marshalling box)
- xxvii) Parallel operation (auto/manual modes) protection and interlock circuits (simulation tests).
- xxviii) Neutral grounding resistor especially for earthing connection and earth resistance.
- xxix) Phase sequence (when transformers have to be paralleled)
- xxx) Sealing of busduct entries through walls (for vapour, dust, water tightness)
- xxxi) Vermin & dust proofness and earth connections of bus ducts
- xxxii) Busduct supports
- xxxiii) Expansion joints and flexible connection of busduct
- xxxiv) Cable terminations
- xxxv) Radiator valves are in open position.
- xxxvi) Provision of caution boards
- xxvi) Magnetic core balance test
- xxvii) Vector group test
- xxviii) Availability of conservator oil level indicator, minimum level marking and drain plug
- xxix) Availability of valves for filtration, drainage and fillings with necessary plugs

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	OISD – STD – 137	Page No. 43
	INSPECTION OF ELECTRICAL EQUIPMENT	

- xxx) Diagram plate in a visible position showing the internal connection, vector diagram and terminal details.
- xxxii) Fire buckets confirming to IS 2546:1974 with the suitable stand.
- xxxii) Checks/ testing as recommended by OEM.

After commissioning the transformer, the following points shall be observed, and recorded:

- i) Oil leakage, if any
- ii) The no-load current
- iii) Alarm, if any, or any relay flag has operated
- iv) Voltage and current on all three phases
- v) Transformer hum or abnormal noise, if any
- vi) Current, voltage, and temperature readings (hourly for 24 hours when transformer is loaded)
- vii) Phase sequence checking (parallel operation)
- viii) Oil & winding temperature

9.2.4 Battery & Charger

The items listed below shall be checked/ ensured:

- i) Battery Bank not exposed to direct sunlight.
- ii) Healthiness of Battery charger Float and Boost charging operations
- iii) Proper ventilation and provision of acid proof flooring in the battery room
- iv) Check battery banks for healthy cell voltage, specific gravity of cells and electrolyte level.
- v) Cell voltage, proper cell connections to maintain correct polarity.
- vi) Corrosion problems especially to terminals, if any.
- vii) Damages of battery container /terminal , if any
- viii) Check for spillage of electrolyte and ensure Battery Room is dry
- ix) Tightness of wiring connections
- x) Healthiness of protection and annunciation circuit of Battery Charger
- xi) Functioning of exhaust fans
- xii) Checks/ testing as recommended by OEM.

9.2.5 Lighting System

The following shall be checked/ tested:

- i) Installation and components meet the requirements set by environmental conditions and conform to approved drawings/specifications
- ii) Ensure no fuses blown and no lamps fused
- iii) Circuit loading and balancing of loads between phases
- iv) Provision of safety features such as:
 - a. Separation sheets
 - b. Fuse base screening
 - c. Door switches & padlocks
- v) Continuity and completeness of wiring connections
- vi) Tightness of wiring connections at the terminals
- vii) Cable terminals
- viii) Ratings of fuses, MCBs, ELCBs and switches
- ix) Insulation resistance between phases and between phases & earth.
- x) Weatherproofing and vermin proofness.
- xi) Earthing connection/earth resistance values
- xii) Size and type of cables, wires and conduits used.
- xiii) Painting (corrosion effect)
- xiv) Functional checks of lighting circuits for operation

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	OISD – STD – 137	Page No. 44
	INSPECTION OF ELECTRICAL EQUIPMENT	

- xv) Operation of emergency lighting circuits and its automatic starting when normal use supply fails (simulation)
- xvi) Condition of lighting fixtures, panels & distribution boards
- xvii) Available illumination levels areawise to confirm operation and safety requirements through field inspection with UIC. Ensure that minimum lux level for normal and emergency lighting is maintained in line with OISD-STD-149.
- xviii) Lighting circuit identifications and markings on light fixtures in the field.
- xix) Checks/ testing as recommended by OEM.

9.2.6 CABLES

The items listed below shall be checked/ ensured:

- i) Identification of the cable with provision of cable route markers and joint location markers and cable tags
- ii) Voltage rating, size and type of the cable (to ensure correct specifications)
- iii) Cable terminals (especially for flameproof, watertight terminations)
- iv) Proper cable support/ clamping arrangement
- v) Continuity of core to confirm no open circuit.
- vi) Proper cable termination, Armour earthing and bonding. In case of single core cables, armour should be earthed at one end only. Trefoil formation should be ensured for single core cables. Conduits and clamps for trefoil & single core cables shall be non-ferrous type.
- vii) Insulation resistance and Polarisation index testing of cables
- viii) Checks/ testing as recommended by OEM.

9.2.7 SPECIAL EQUIPMENTS

Special equipment like UPS, VFDs, Battery chargers, Generator AVR/ DVRs, Impressed current Cathodic protection (ICCP) panels, PLCs, PLCCs, GIS, ECS etc. should be installed and commissioned strictly as per OEM check-lists and guidelines.

	OISD – STD – 137	Page No. 45
	INSPECTION OF ELECTRICAL EQUIPMENT	

10.0 REFERENCES

The following codes, standards and publications have either been referred to or used in the preparation of this standard, and the same shall be read in conjunction with this standard:

- i) CEA (Measures relating to Safety and Electric Supply) Regulation, 2010
- ii) BIS (Bureau of Indian Standards)- IS:2705, IS:3156, IS:3043, IS:1554, IS:2309, IS:5216, IS:335, IS:266, IS:900, IS:3156, IS:10029, IS:1255, IS:732, IS:5572, IS:2148, IS:7389, IS:3072, IS:4770, IS:10118, IS:1866, IS:2516, IS:5124,
- iii) National Electrical Code, 1985
- iv) Model code of safe practice Electrical-Part 1, 1974 (Institute of Petroleum publication)
- v) Guide for Inspection of Refinery Equipment-Chapter XIV-Electrical System-(API-Publication) 1976
- vi) OISD-STD-105 'Work Permit System'
- vii) British Standard Code of Practice BS-6651, CP-1013
- viii) Regulations for the electrical equipment of buildings. Fifteenth edition 1976 (Institution of Electrical Engineers - IEE's publication)
- ix) Oil Mine Regulation, 1984
- x) Petroleum Rule, 2002
- xi) Asian Electricity Journal, Oct 96.
- xii) Fault finding solutions article - Megger
- xiii) OSHAS: CFR 1910 -147, 331
- xiv) IEEE Std. 43-2000: IEEE Recommended Practice for Testing Insulation Resistance of Rotating Machinery
- xv) NFPA 70B: Recommended Practice for Electrical Equipment Maintenance
- xvi) CBIP Manual on EHV Substation Equipment Maintenance (Publication No. 294)
- xvii) Maintenance Testing Specifications for Electric Power Distribution Equipment and Systems - NETA
- xviii) ANSI/IEEE C57.104: IEEE Guide for the Detection and Determination of Generated Gases in Oil-Immersed Transformers and Their Relation to the Serviceability of the Equipment.
- xix) IEEE 400.2- 2004: IEEE Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)
- xx) IS 7098 -Part 2: Crosslinked polyethylene insulated thermoplastic sheathed cables - specification
- xxi) IEEE -400.3-2006: IEEE Guide for Partial Discharge Testing of Shielded Power Cable Systems in a Field Environment
- xxii) IS 11322: Method for Partial Discharge Measurement in Instrument Transformers
- xxiii) IS 1866: Code of practice for electrical maintenance and supervision of mineral insulating oil in equipment.
- xxiv) CPWD, General Specifications on Electrical Works: Part-I(Internal) and Part-IV (Sub-station), 2013

	OISD – STD – 137	Page No. 46
	INSPECTION OF ELECTRICAL EQUIPMENT	

ANNEXURE I

HISTORY CARDS

ELECTRO MAGNETIC RELAY INSPECTION AND TEST RECORD	
CIRCUIT LOCATION _____ CIRCUIT DESIGNATION _____	
RELAY _____ PHASE OR LOCATION IN PANEL _____	
MANUFACTURER _____ TYPE _____ STYLE NO. _____ FUNCTION _____	
C.T. RATIO _____ P.T. RATIO _____	
TAP RANGE _____ INSTANTANEOUS RANGE _____	
TIME OPERATIONS INDICATOR _____ AND INSTANTANEOUS INDICATOR _____ AMP	
Relay Inspection date	
Cover	
- Glass tight - Glass clean - Gasket tight	
Relay condition	
- Dust - Rust - Iron filings	
Bearings sticky	
Shorting bars in place	
Connections tight	
Contacts	
- Dirty - Pitted	
Operations indicator	
- Armature free - Target free	
Instantaneous attachment	
- Armature free - Target free	
Disc rubbing inspection	
Magnet condition	
Auxiliary contact condition	
- NO - NC	
Auxiliary coil pick-up for contact multiplier	
CT terminal condition / tightness checking	
Disc movement starting current	
Disc Rotating spring condition / availability	
Inst Trip operation	
Time contact make spring adjustment	
Flag operation	
Latch operation	
Moving Contact Spring condition	

Note: For Numerical relay testing format by OEM to be followed. Also, it should be ensured that the 'relay internal failure signal' is configured for alarm.

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	OISD – STD – 137	Page No. 47
	INSPECTION OF ELECTRICAL EQUIPMENT	

CABLE MAINTENANCE CARD					
Card No. _____					
Make	Type	Year of Installation	Tag No.	Size	
Voltage	Current	Length	Initial No. of Joints	Cable route line marker	
Date	Type of fault	Fault location	Distance From length (m)	Job carried out	Executed by

		MOTOR DETAILS :		PLANT :	
PURCHASE ORDER :		MFGR. REF. :		TAG NO. :	
DATE OF COMMISSIONING :		DRG. NO. :		SERVICE :	
MOTOR TYPE :		HP / KW		STARTING METHOD	
SERIAL NO.		RPM		INSUL. CLASS	
MAKE		VOLTS		TEMP. CLASS	
MOUNTING _____ FOOT / FLANGE MOUNTED		PHASE _____ FREQUENCY _____		DUTY _____	
FRAME SIZE _____		FLC _____ NLC _____		TYPE OF PROTECTION _____	
WINDING CONNECTION _____ STAR / DELTA _____		POWER FACTOR _____		TYPE OF ENCLOSURE _____	
		EFFICIENCY _____		DIRECTION OF ROTATION (VIEWED FROM NDE) : _____ CW / CCW	
STARTING TIME - UNCOUPLED _____ SECONDS		WINDING RESISTANCE PER PHASE _____			
STARTING TIME - COUPLED _____ SECONDS		WINDING INSUL. RESISTANCE _____			
STARTING CURRENT _____ X FLC		GD ² (MOTOR) _____			
STARTING TORQUE _____ X FLT		SLIP _____ % AT _____ % LOAD		WEIGHT _____ KG	
VOLTAGE AT STARTING _____		SOUND LEVEL _____		ROTAR WEIGHT _____ KG	
DE BEARING _____		DE ON LINE GREASING FACILITY _____ YES / NO		FOR OIL LUBRICATED BEARINGS _____	
NDE BEARING _____		NDE ON LINE GREASING FACILITY _____ YES / NO		TYPE OF OIL _____	
TYPE OF GREASE _____				VISCOSITY OF OIL _____	
DE GREASING INTERVAL _____ WH		GREASE QUANTITY _____ GRAMS		COUPLING TYPE _____	
NDE GREASING INTERVAL _____ WH		GREASE QUANTITY _____ GRAMS		COUPLING SERIES _____	
MOTOR SPACE HEATER _____ YES / NO		SPARE MOTOR AVAILABLE _____ YES/NO		BASE PLATE DIMENSIONS. SEE ALSO DRG.	
VOLTS _____		SPARE MOTOR DETAILS _____			
WATTS _____					

[illegible]

[illegible]

	STARTER & CABLE DETAILS	PLANT :
MCC NO.	MCC P.O. NO.	MAKE DATE OF COMMISSIONING :
MODULE NO.	MODULE RATING	MODULE DRG. NO.
INCOMING PLUG IN CONTACTS RATING	TYPE	MAKE
OUTGOING PLUG IN CONTACTS RATING	TYPE	MAKE
SWITCH FUSE UNIT RATING	TYPE	MAKE
POWER FUSE RATING	TYPE	MAKE
POWER CONTACTOR RATING	TYPE	MAKE
CONTROL FUSE RATING	TYPE	MAKE CONTROL VOLTAGE
CONTROL SWITCH RATING	TYPE	MAKE
<u>THERMAL OIL RELAY</u> - TYPE : - RANGE : - SETTING : - MAKE :	CONTACTOR MOUNTED / INDIVIDUAL MOUNTED WITH / WITHOUT SPP DIRECT / CT / OPERATED	IF CT OPERATED CT RATIO VOLTAGE RATING TYPE VA ACCURACY CLASS MAKE
<u>AMMETER TYPE</u> - TAUT BAND TYPE/ORDINARY RANGE WITH SUPPRESSED SCALE ACCURACY CLASS MAKE	MOUNTING DIMENSIONS DIRECT / CT / OPERATED	IF CT OPERATED CT RATIO VOLTAGE RATING TYPE VA ACCURACY CLASS MAKE
<u>POWER JN. BOX</u> NO. OF ENTRIES ENTRY SIZE JN BOX CONNECTION-WITH/WITHOUT CONNECTING BLOCK	<u>PUSH BUTTON STATION</u> NO. OF ENTRIES ENTRY SIZE	<u>INTERLOCKS DETAILS</u>
POWER CABLE - FROM MCC TO JN. BOX SIZE POWER CABLE - FROM JN. BOX TO MOTOR SIZE CONTROL CABLE SIZE SPACE HEATER CABLE SIZE POWER CABLE - INITIAL I.R. VALUE	LENGTH (Mts.) LENGTH (Mts.) LENGTH (Mts.) LENGTH (Mts.) CONTROL CABLE INITIAL I.R. VALUE	TAG. NO. TAG. NO. TAG. NO. SPACE HEATER CABLE INITIAL I.R. VALUE

EARTH RESISTANCE RECORD CARD

SUBSTATION NO.-
GANTRY / TANK NO.-
NO. OF EARTH PITS-

EARTH CONDUCTOR DETAILS

[illegible]

	OISD – STD – 137	Page No. 52
	INSPECTION OF ELECTRICAL EQUIPMENT	

BATTERY BANK HISTORY CARD

LOCATION	
SERVICE	
MAKE	
TYPE	
MODEL	
CAPACITY	
NO.OF CELLS	
TOTAL VOLTAGE	
YEAR OF COMMISSIONING	
SPECIFIC GRAVITY	

MONTH	DATE	AV. CELL VOLTS	TOTAL VOLTAGE	REMARKS	SIGN
JANUARY					
FEBUARY					
MARCH					
APRIL					
MAY					
JUNE					
JULY					
AUGUST					
SEPTEMBER					
OCTOBER					
NOVEMBER					
DECEMBER					

Note: For individual Battery Bank, monthly data sheet should be maintained.

	OISD – STD – 137	Page No. 53
	INSPECTION OF ELECTRICAL EQUIPMENT	

ANNEXURE II**FAILURE ANALYSIS REPORT**

INSTALLATION.....UNIT
 EQUIPMENT NAME:EQUIPMENT TAG NO.:
 DATE

- | | |
|---|--------------------------|
| 1. Present Failure/Abnormality | Last Failure/Abnormality |
| Date: | Date: |
| Nature: | Nature: |
| 2. Component Damage/Abnormality | Cause of Damage. |
| 3. Principal Cause of Failure: Facts / Observations | |
| 4. Reasons – why it happened. | |
| 5. Repair/Replacement Carried out | |
| 6. Steps taken to prevent repetition | |
| 7. Lessons learnt/ Recommendations | |

Signature/name/Designation
 (of the Officer preparing the Report)

	OISD – STD – 137	Page No. 54
	INSPECTION OF ELECTRICAL EQUIPMENT	

ANNEXURE III**DETAILS OF DESIGNATED PERSONNEL**

INSTALLATION

DATE:.....

Sl.No.	Name	Designation	Technical Competency	Clause No.of CEA Regulation 2010 For which Designated	The work responsibility entrusted to designated person	Signature of person in Acceptance
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Signature/name/Designation
(of the Officer preparing the Report)

	OISD – STD – 137	Page No. 55
	INSPECTION OF ELECTRICAL EQUIPMENT	

ANNEXURE IV**CABLE FAULT DETECTION****GENERAL**

When at some local point in a cable, insulation has deteriorated to a degree that a breakdown occurs allowing a surge of current to ground, the cable is referred to as a faulted cable. The use of particular fault finding equipment depends on the factors such as type of fault, length of cable, location of cable, cable construction etc.

Broad guidelines for methods and equipment necessary**The methods and equipment necessary are:**

- | | |
|--|---|
| • Diagnosis
(Fault resistance and loop test) | - Ohm meter
- Insulation Tester |
| • Pre-location | - High voltage Bridge or High Resistance fault locator
- Impulse current Equipment
- Pulse reflection method (pulse echo) |
| • Fault Conditioning
(Sometimes necessary before pre-location to reduce fault resistance) | - Burn down set |
| • Pin-pointing | - Surge Generator
- Surge detector (electromagnetic & acoustic signals) |
| • Route tracing | - Audio Frequency Set |
| • Fault Confirmation | - Acoustic Detector
- Audio Frequency set occasionally. |
| • All above functions | - Combination Sets of portable equipment for full fault location capability. |

Type of fault:

In general, faults can be said to be open or short circuits, However, in practice, many conditions can arise i.e. a break may not be “clean”- it may have resistance across it or, more usually, to the sheath or earth. A contact fault may be “hard down” or exhibit hundreds of ohms, kilo-ohms or even megohms. Also faults may occur on one or more of the cores of conductors.

Fault locating procedure:

- Fault Indication
- Disconnecting and Earthing
- Fault Analyses and Insulation Test
- Cable Fault Prelocation
- Cable Route Tracing
- Precise Cable Fault Location (Pinpointing)
- Cable Identification
- Fault Marking and Repair

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	OISD – STD – 137	Page No. 56
	INSPECTION OF ELECTRICAL EQUIPMENT	

▪ Cable Testing and Diagnosis

(a) Contact fault:

Pulse Echo Set if fault resistance is suitable. If not, H.V. Or High Resistance bridge or Impulse current set can be used.

(b) Flashing Fault:

Impulse Current Set

Choice of instruments

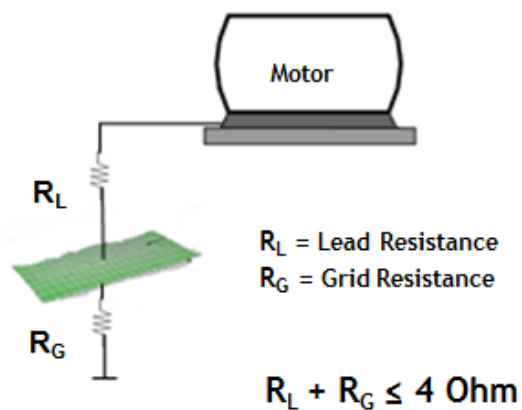
- For XLPE Insulated cable, a full set of equipment incorporating Impulse current equipment will be required, because it is impossible to burn down faults on this type of cable and pulse echo is, therefore, not possible for contact faults.
- For long cables, say up to 15 Km presents few problems in pre-location but Pin-pointing equipment is essential because even a low-pre-location error results in a wide search area.
- For short route lengths a rough “one Instrument” approach may be successful. For Instance, assuming pre-location is possible, the application of modern-powerful Surge generator to test a few hundred meters of cable and careful patrolling of the route often gives a “spot-on” location within a short time.

	OISD – STD – 137	Page No. 57
	INSPECTION OF ELECTRICAL EQUIPMENT	

ANNEXURE-V

TYPICAL DIAGRAMS OF RESISTANCES TO GENERAL MASS OF EARTH

✓ 4 Ohms for Electrical Systems



✓ 7 Ohms for Storage Tanks

